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Nomani et al.

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(54) **SYSTEMS AND METHODS FOR GROUP MESSAGING USING BLOCKCHAIN-BASED SECURE KEY EXCHANGE**

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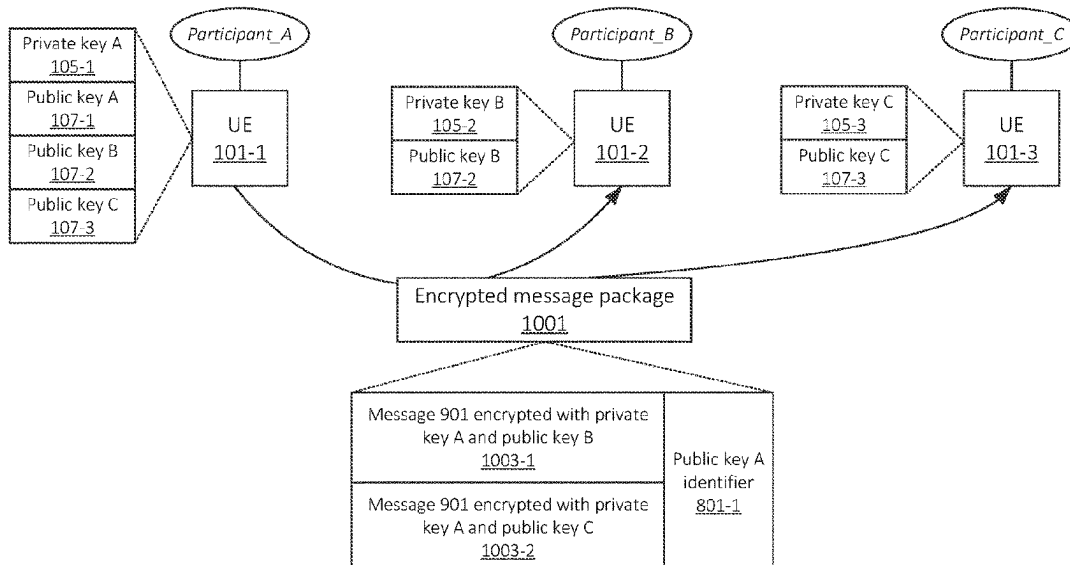
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CPC H04L 63/0442; H04L 9/50; H04L 63/061;
H04L 9/006; H04L 9/0861; H04L 9/14;
H04L 63/12

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(57) **ABSTRACT**

A system described herein provide for the secure maintaining and providing of information, such as public keys used in Public Key Infrastructure (“PKI”) techniques or other techniques, using a secure distributed ledger (e.g., “blockchain”) system. A first entity may generate an asymmetric key pair that includes a public key and a private key, and may provide the public key, along with a key identifier, to a blockchain system for retrieval by multiple other entities. For example, the entities may be engaged in a secure group communication session, in which communications are encrypted and may be decrypted using one or more keys, including the public key. The other entities may identify the public key, as recorded to the blockchain system, based on the key identifier. Participants may be added to or removed from the secure group communication session.

20 Claims, 23 Drawing Sheets



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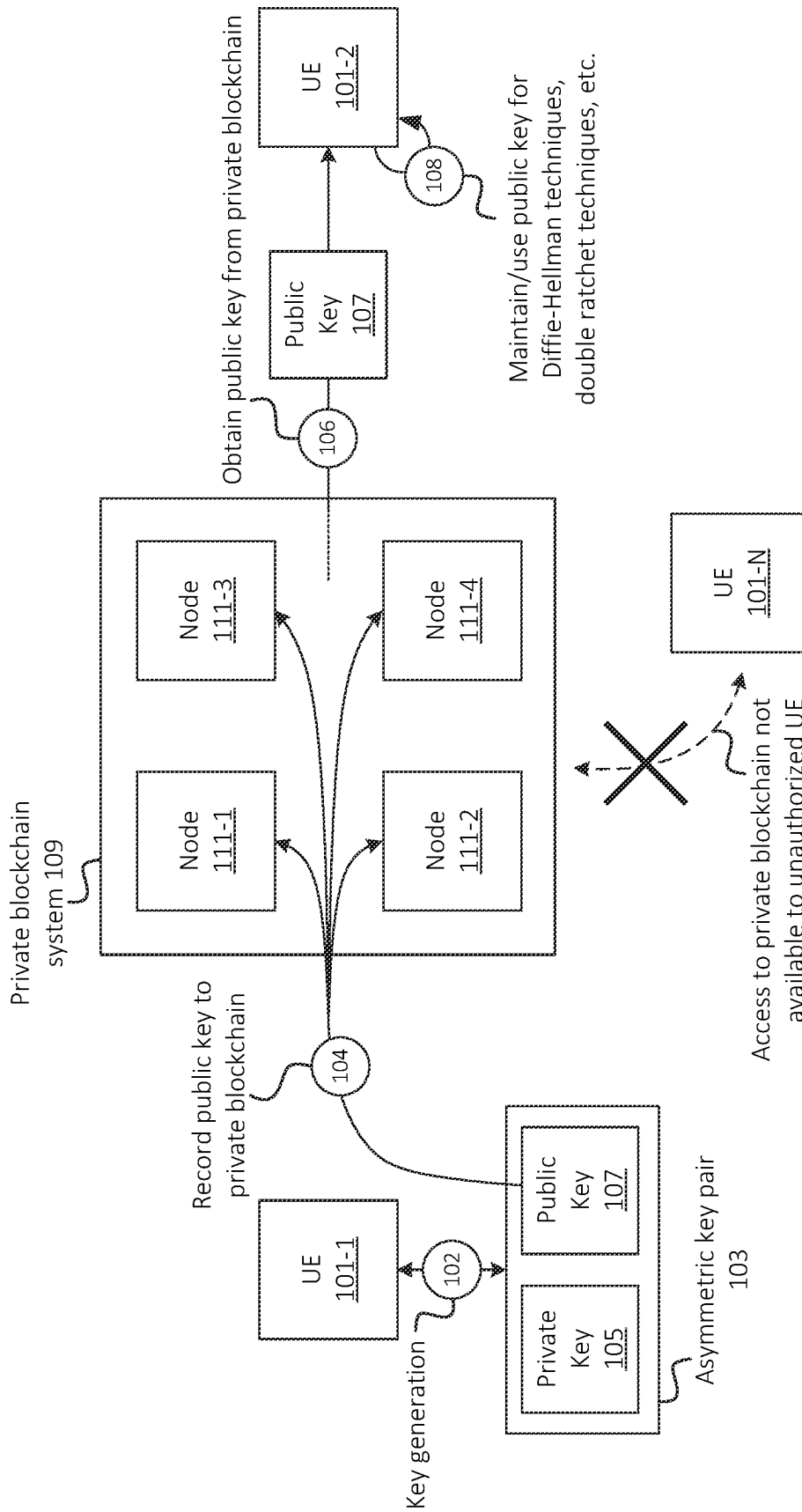


FIG. 1

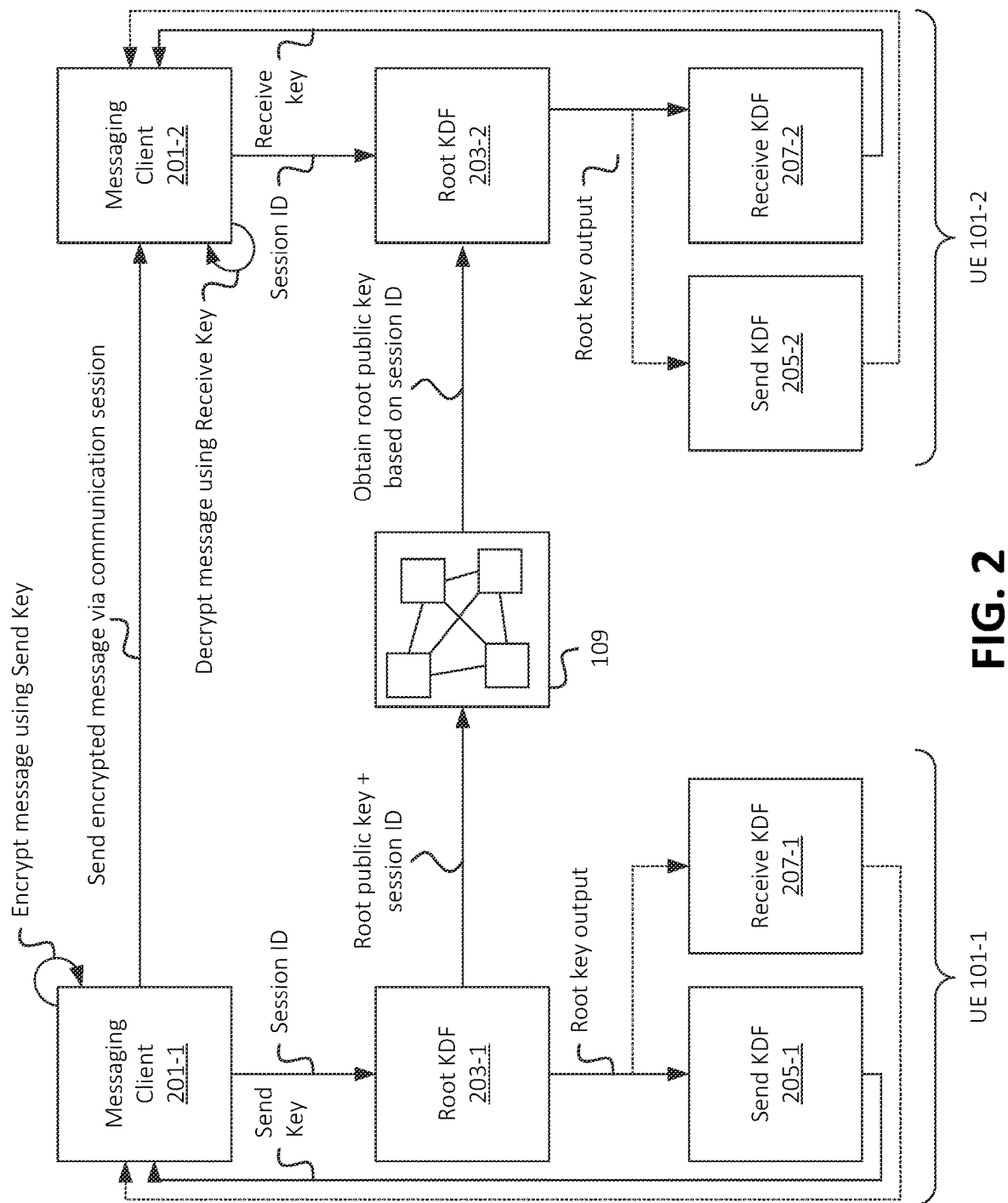


FIG. 2

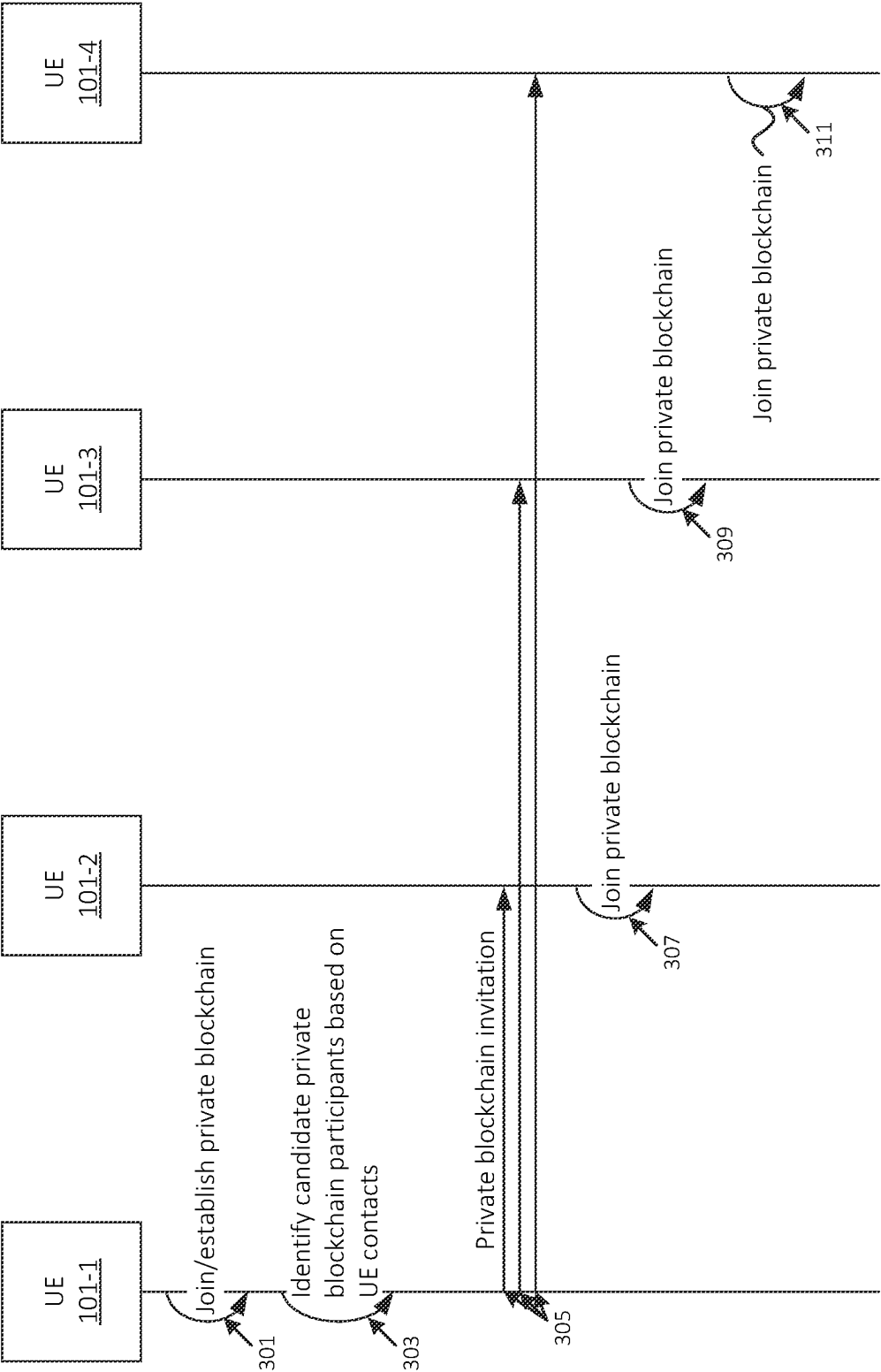


FIG. 3

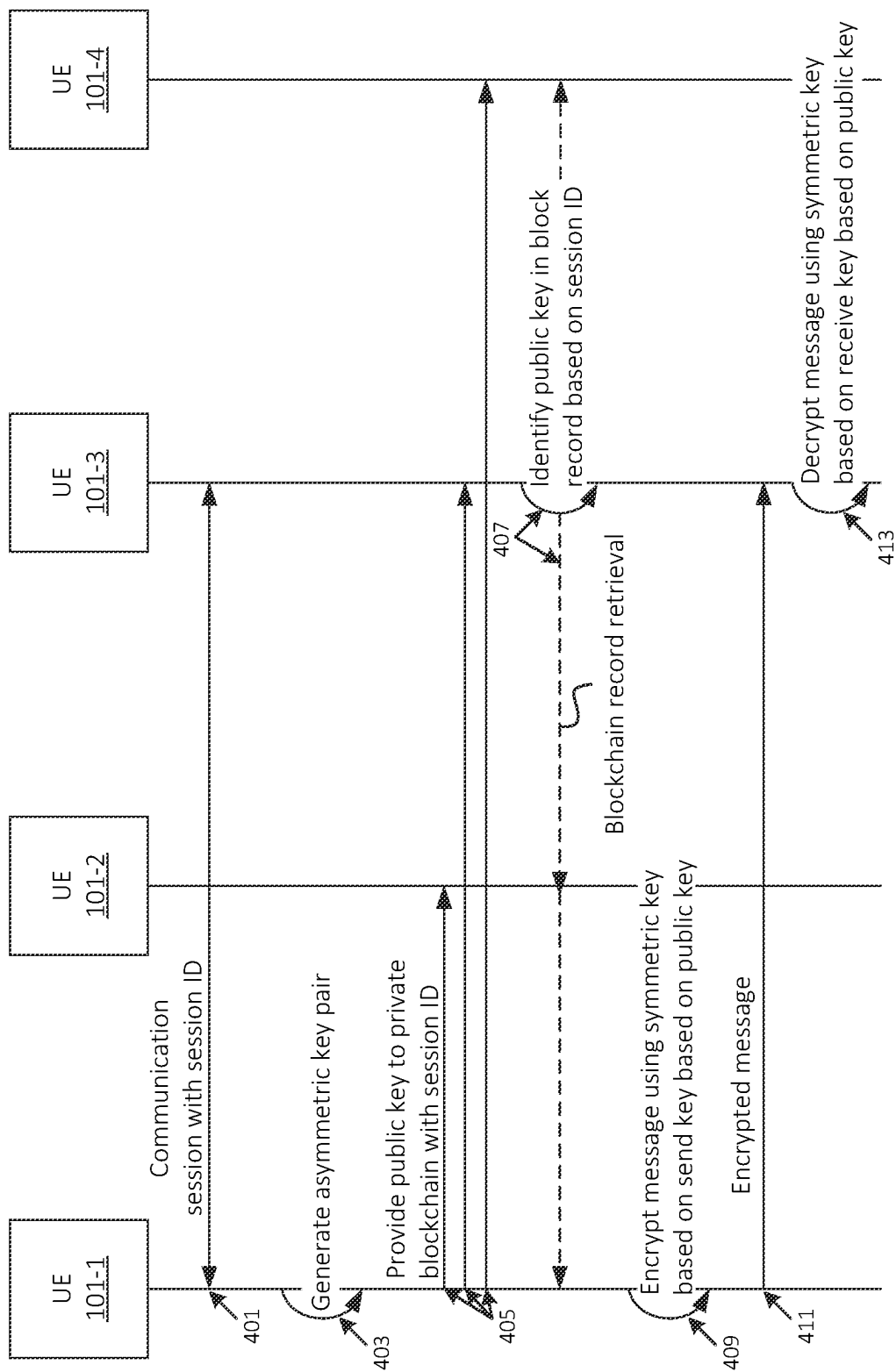
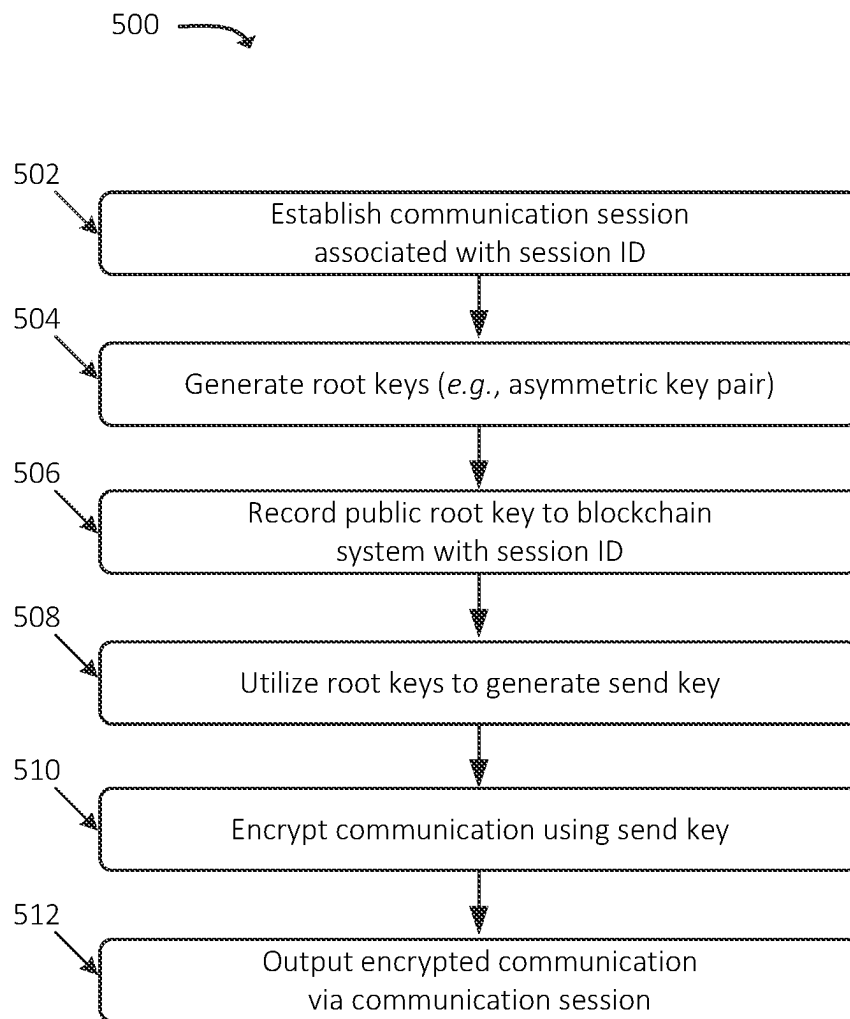
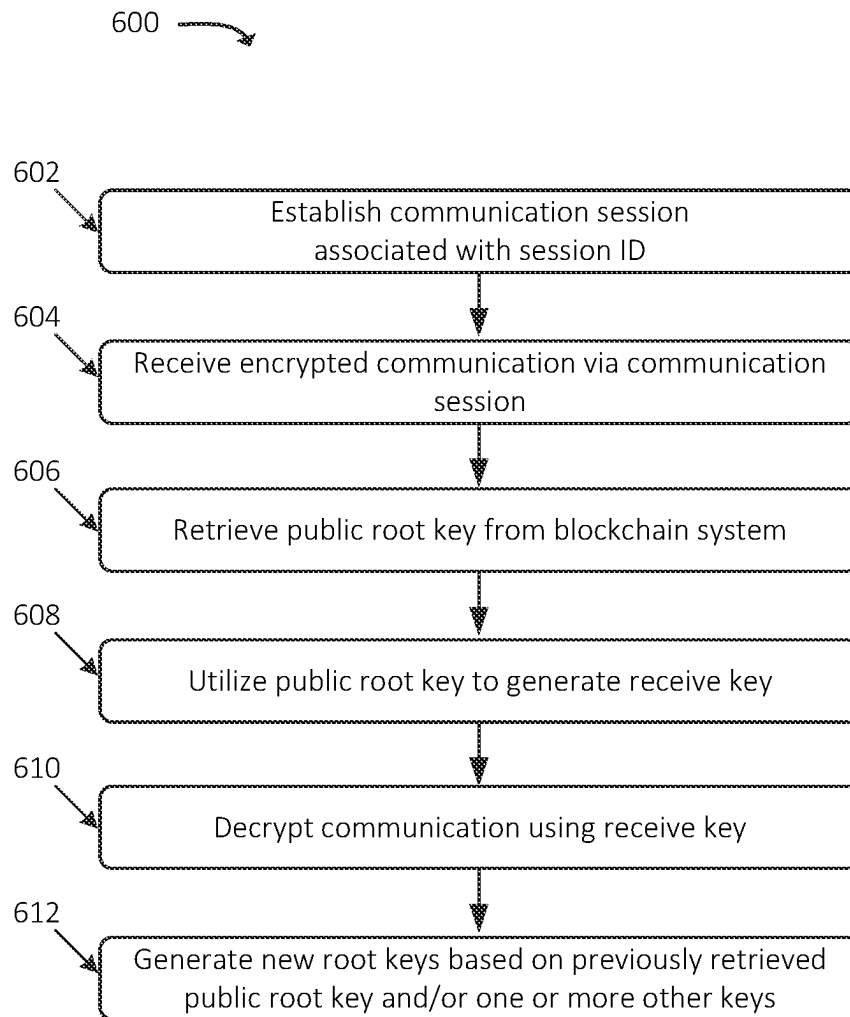


FIG. 4

**FIG. 5**

**FIG. 6**

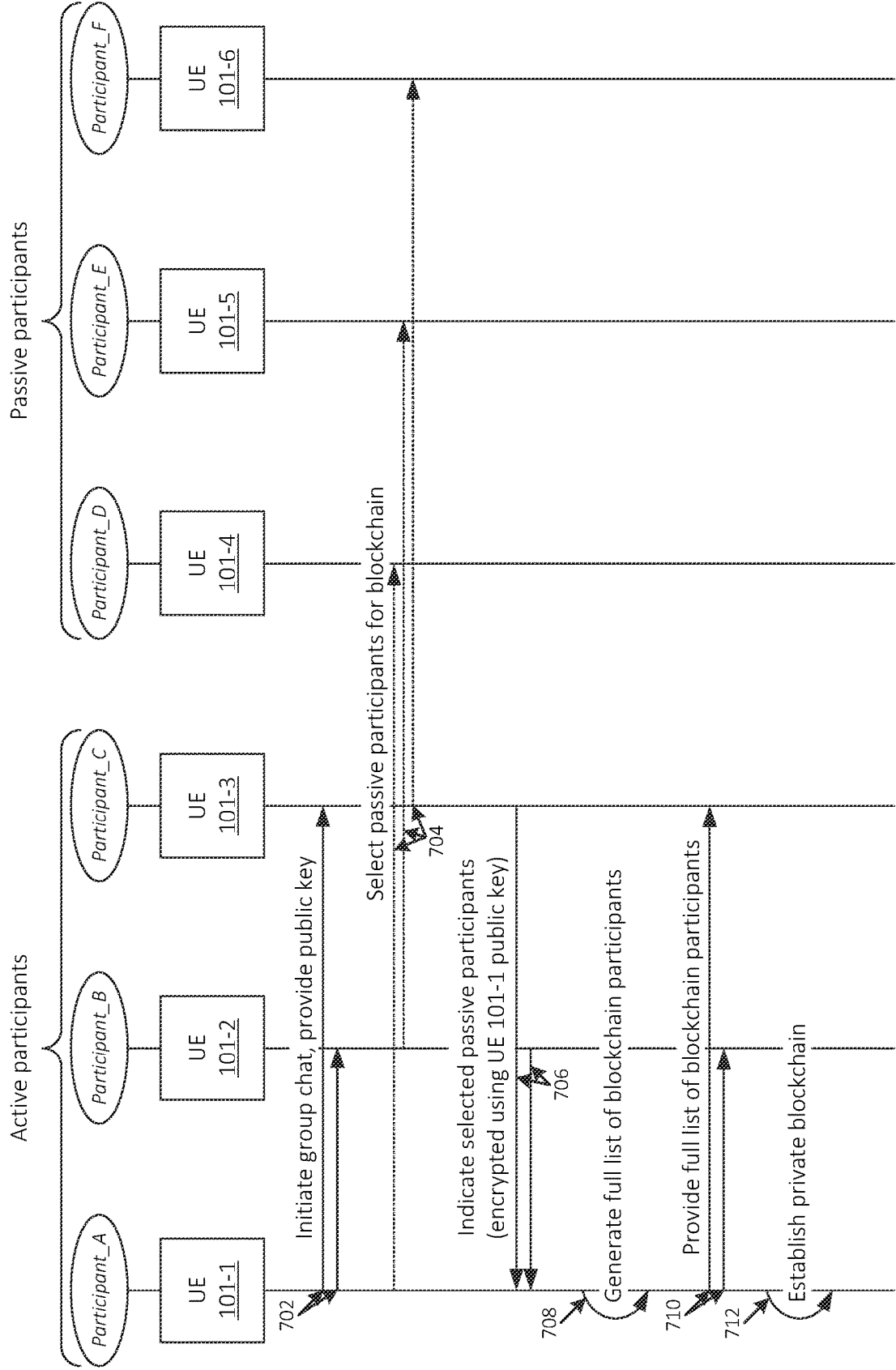


FIG. 7

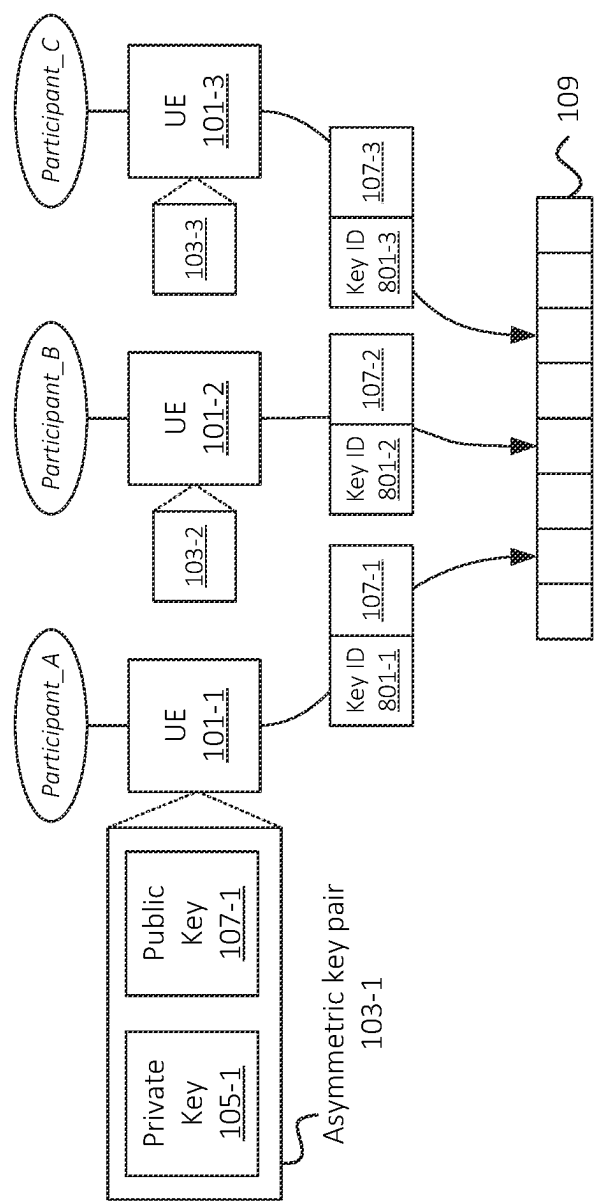


FIG. 8

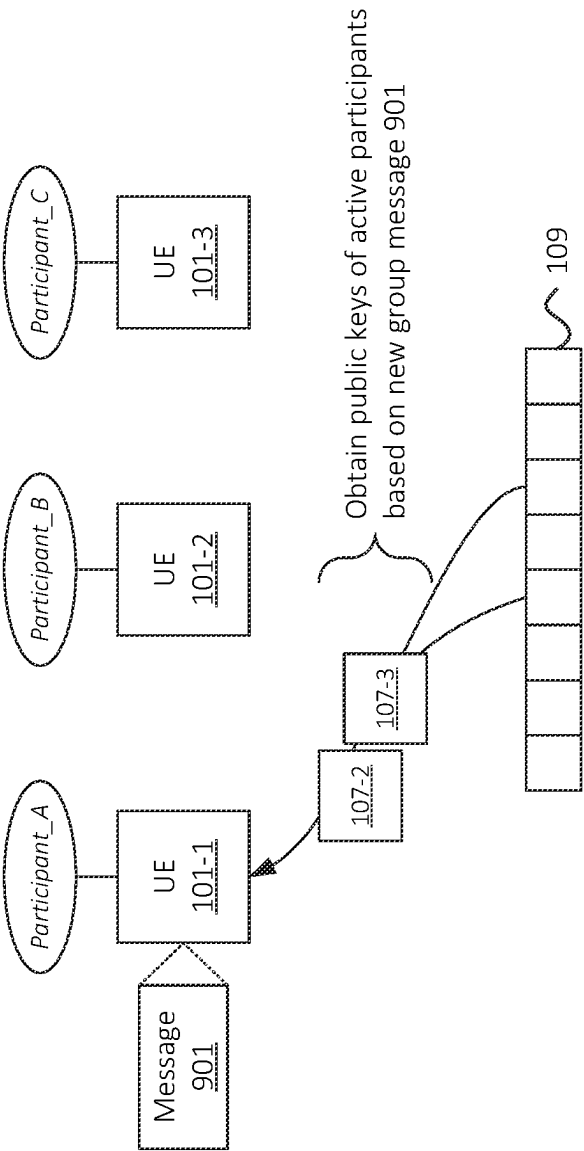


FIG. 9

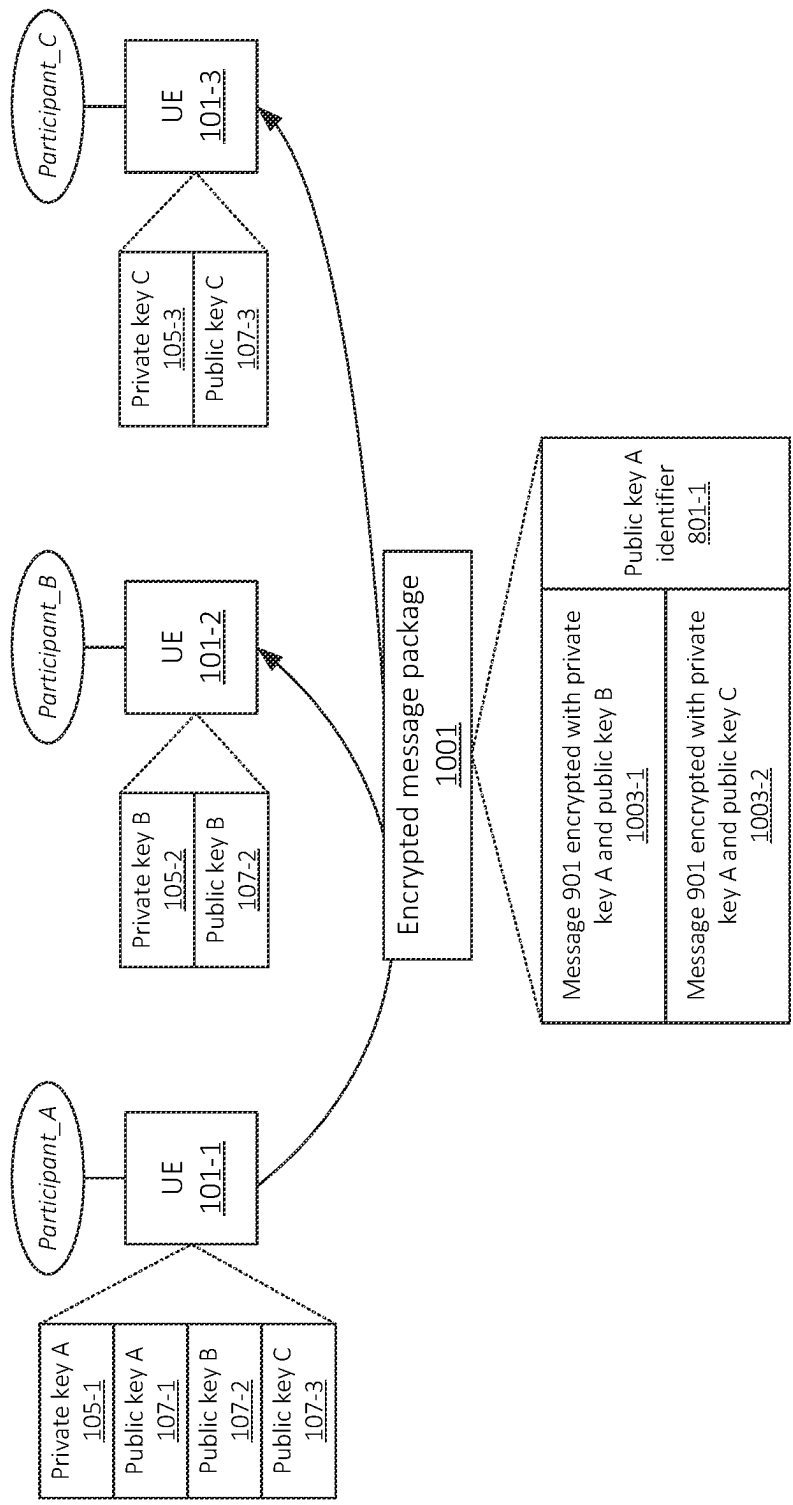


FIG. 10

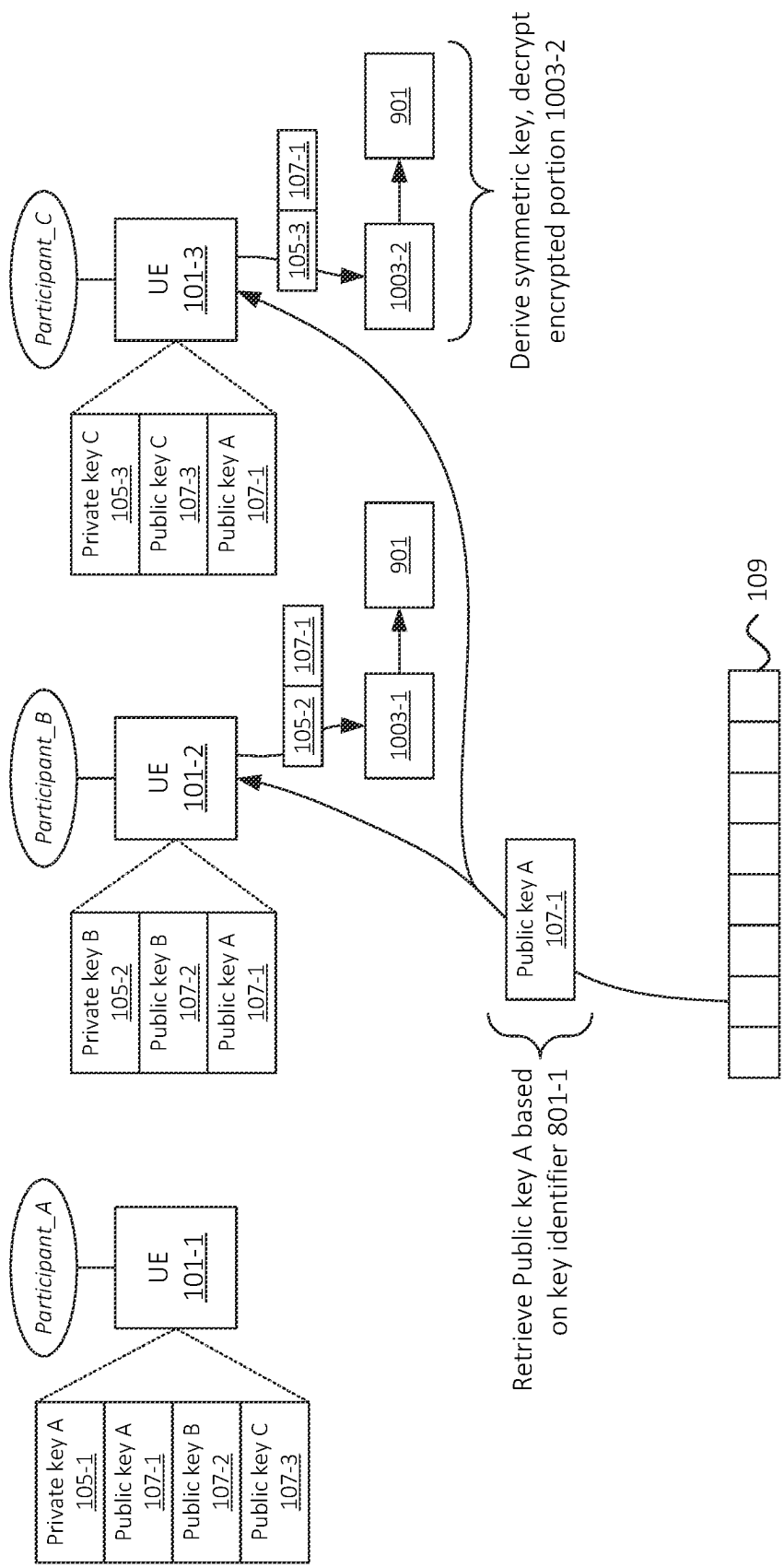


FIG. 11

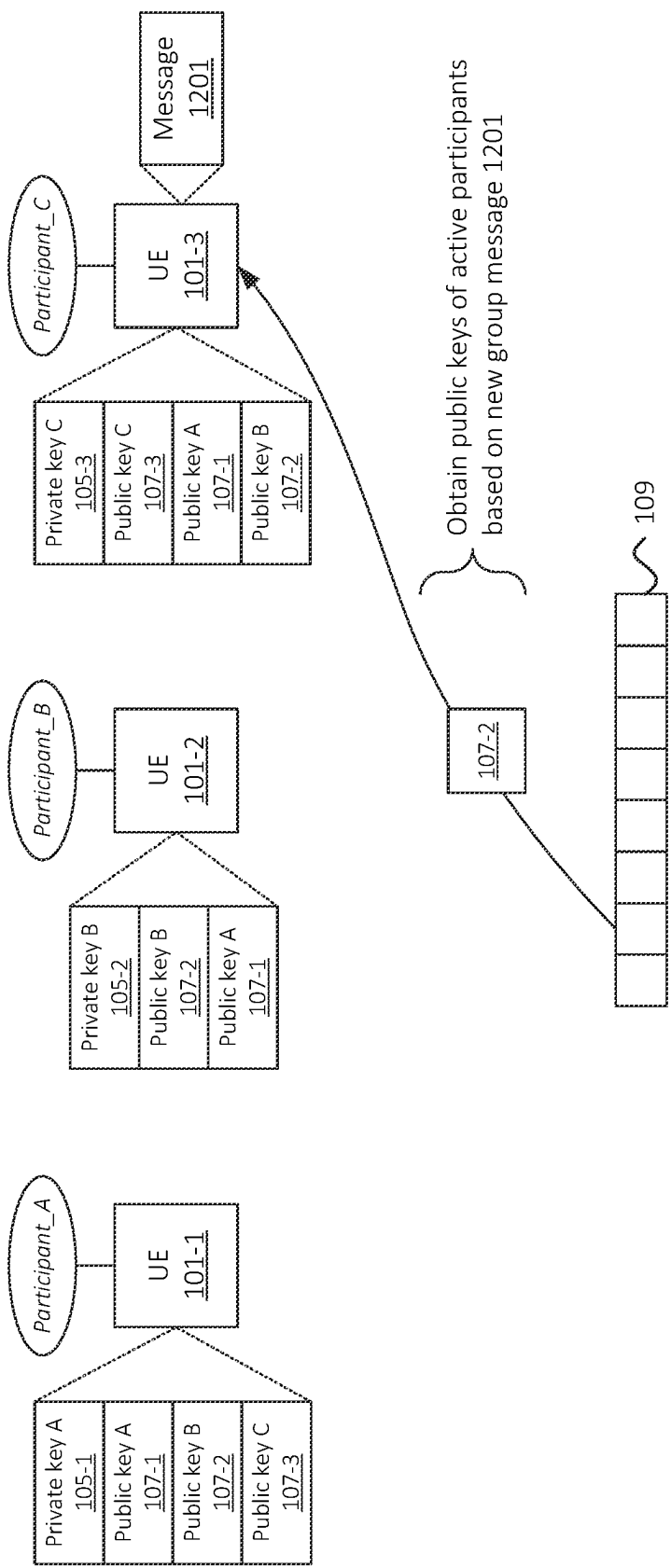


FIG. 12

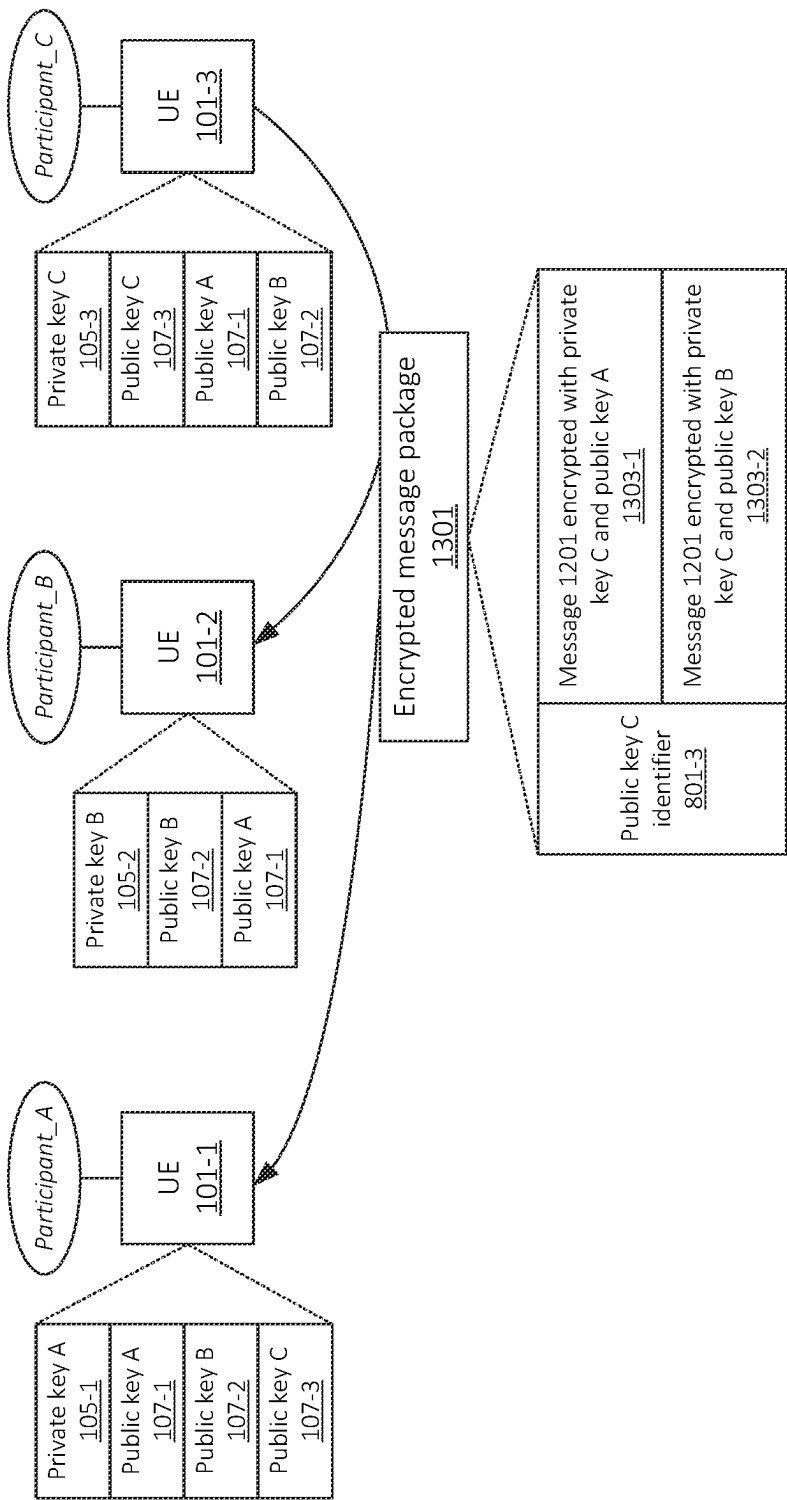


FIG. 13

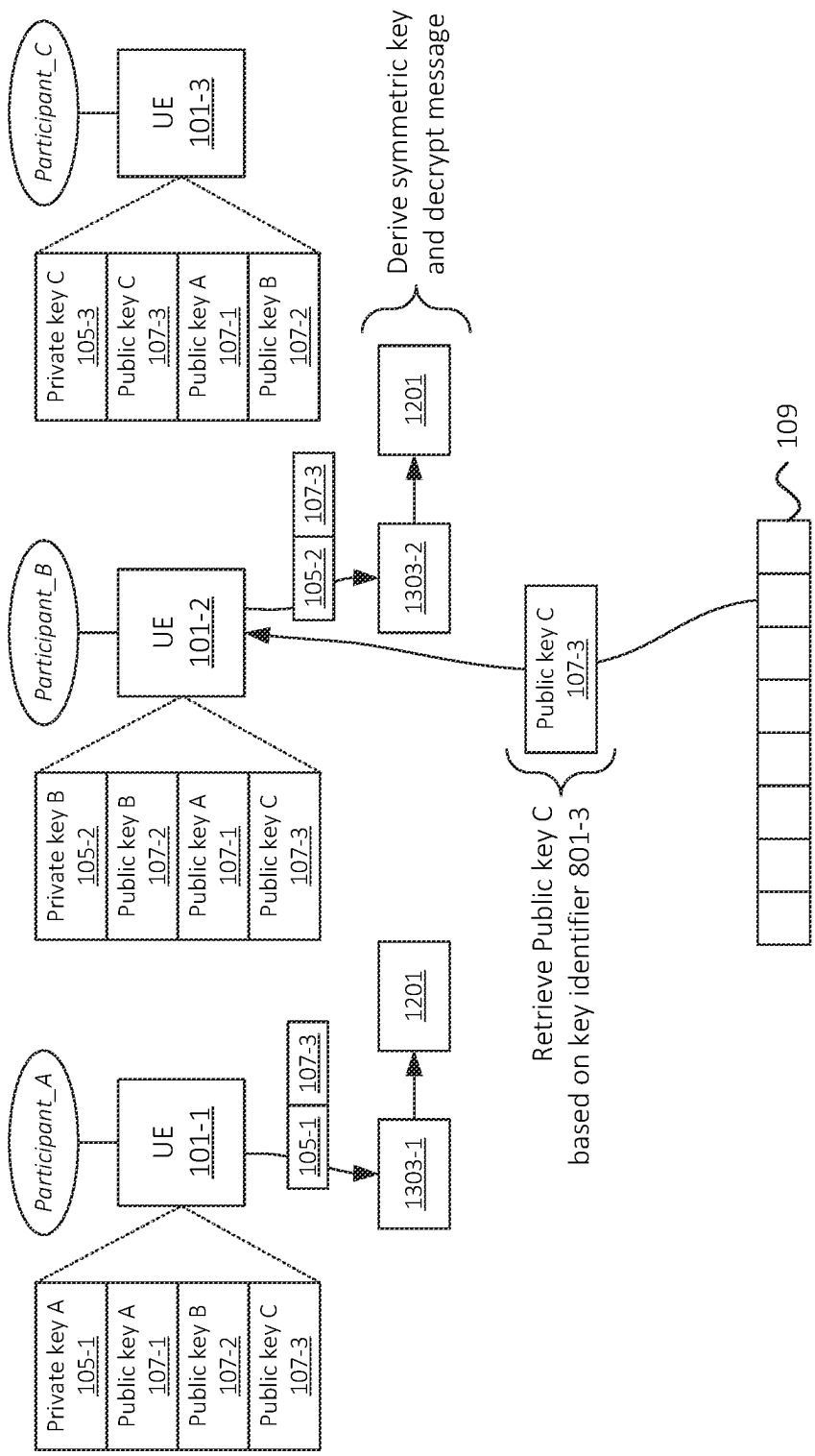


FIG. 14

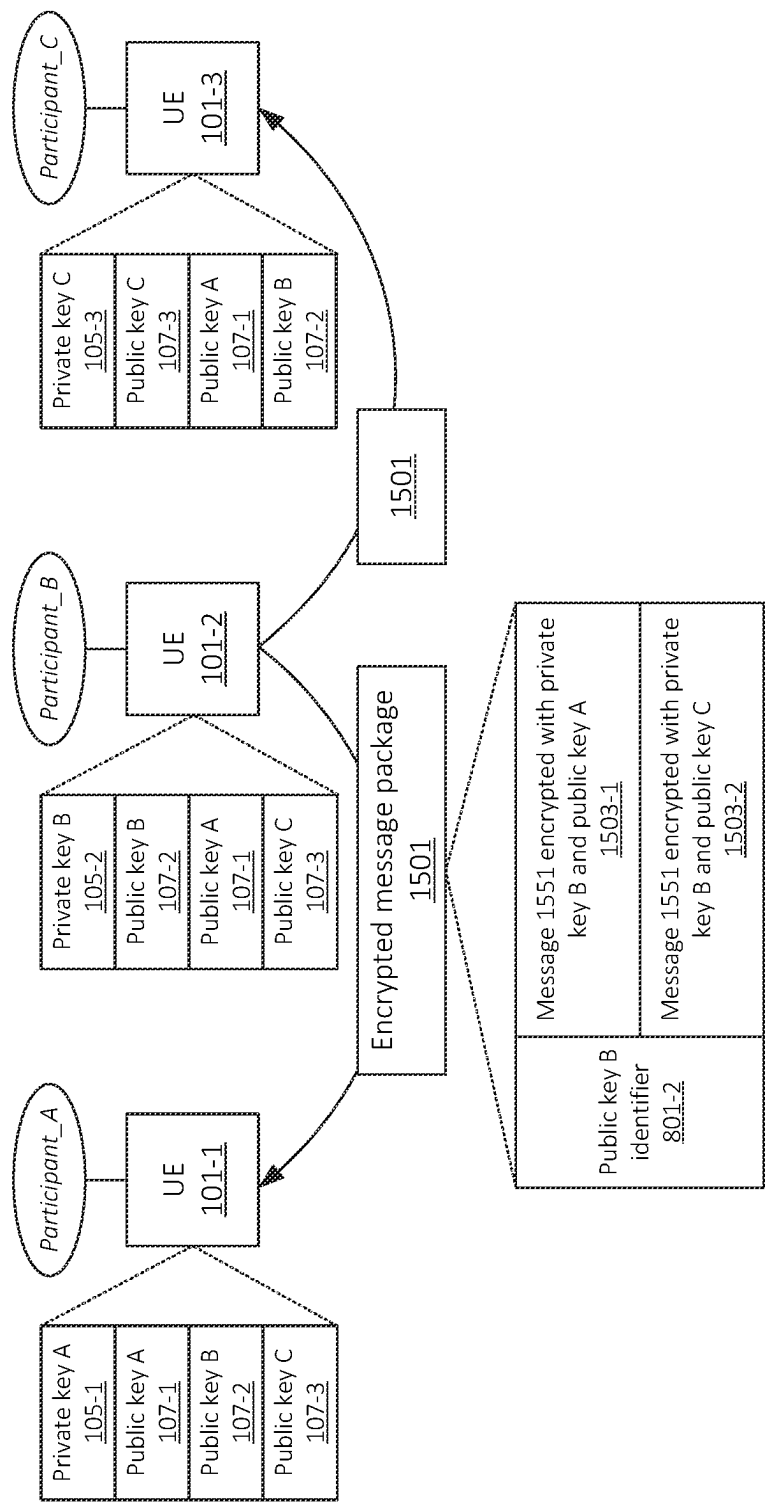


FIG. 15

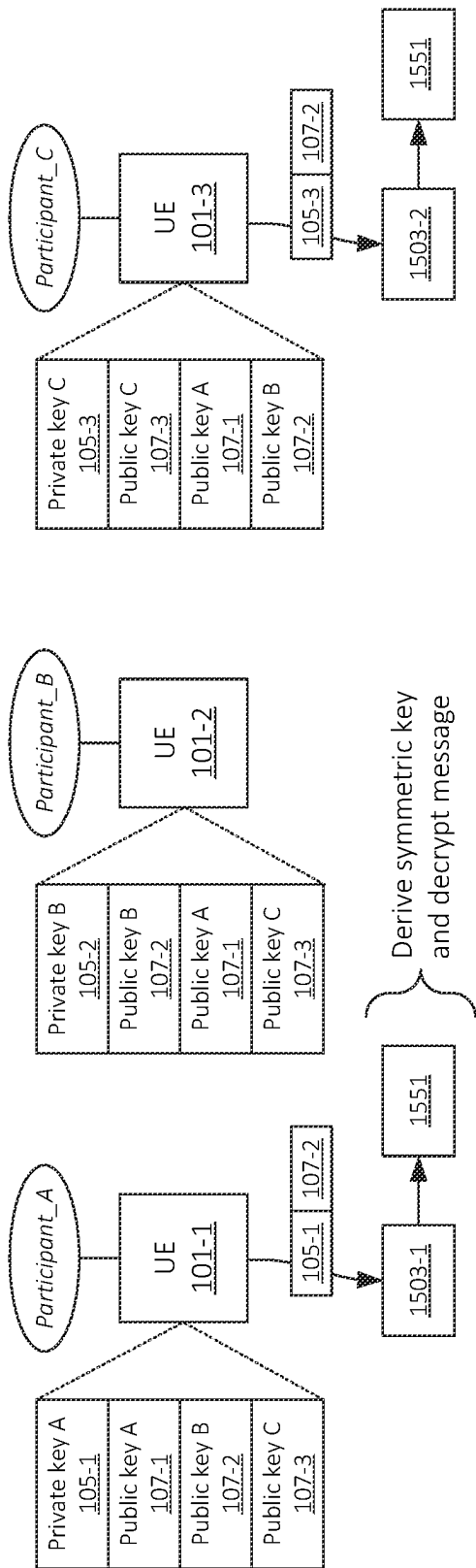


FIG. 16

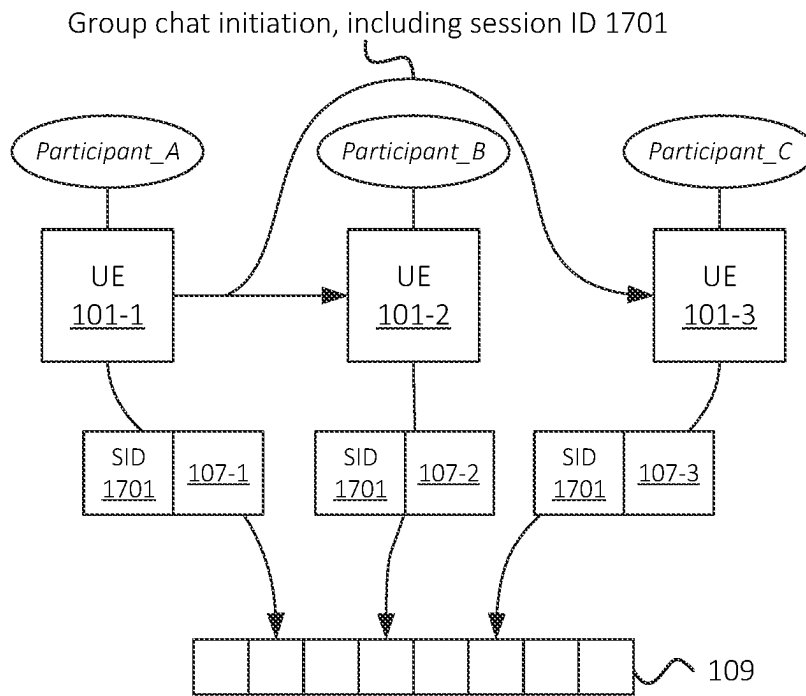


FIG. 17

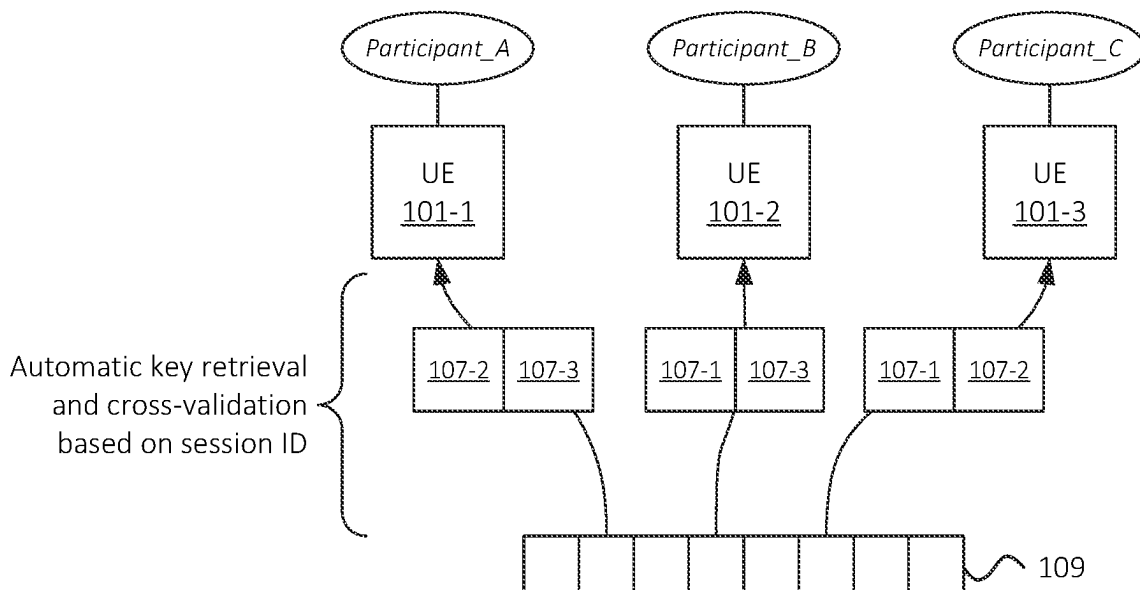


FIG. 18

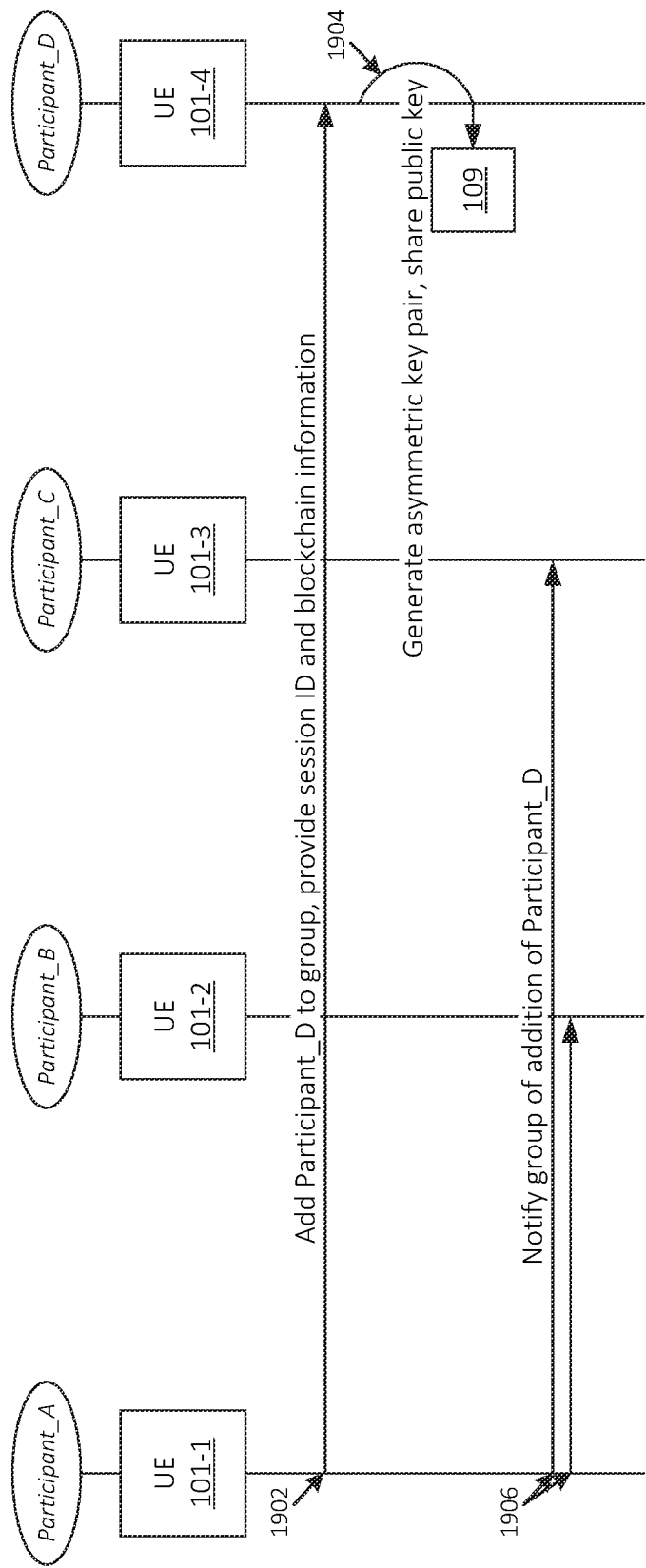


FIG. 19

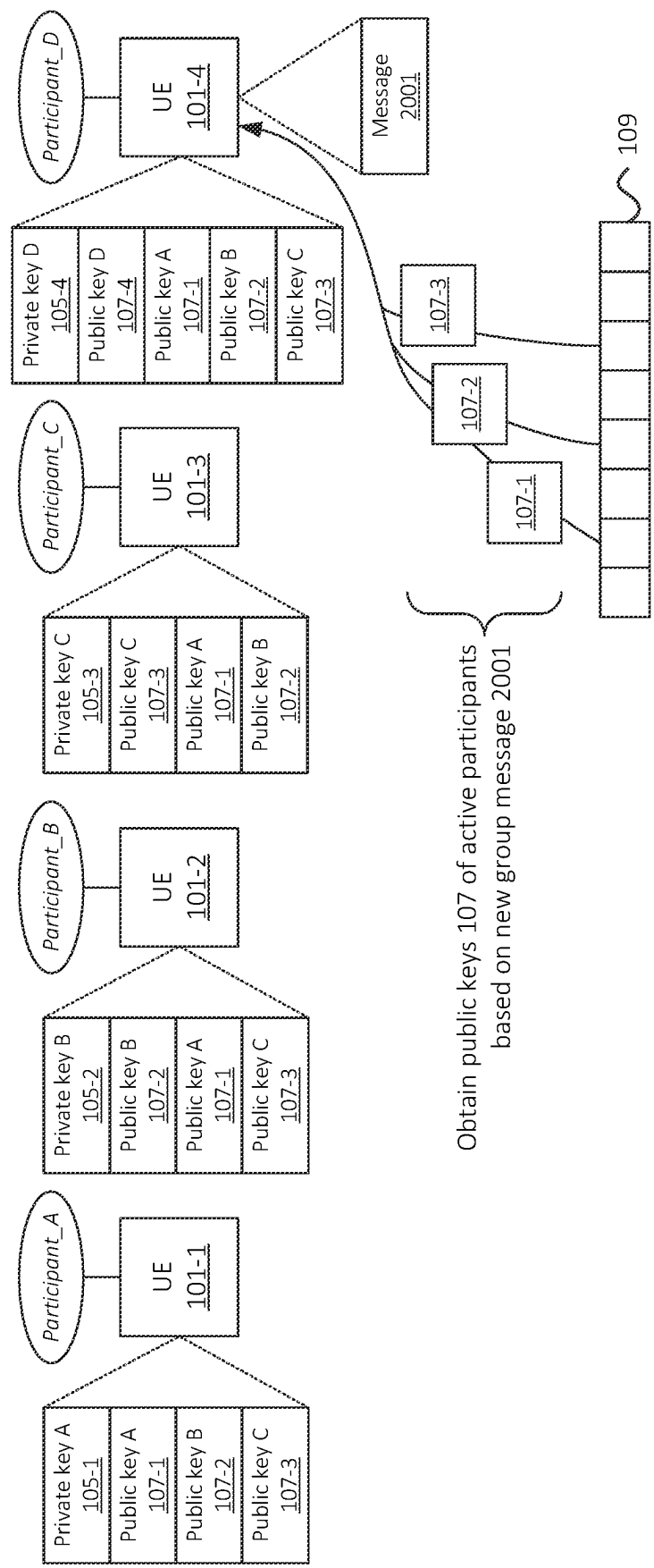


FIG. 20

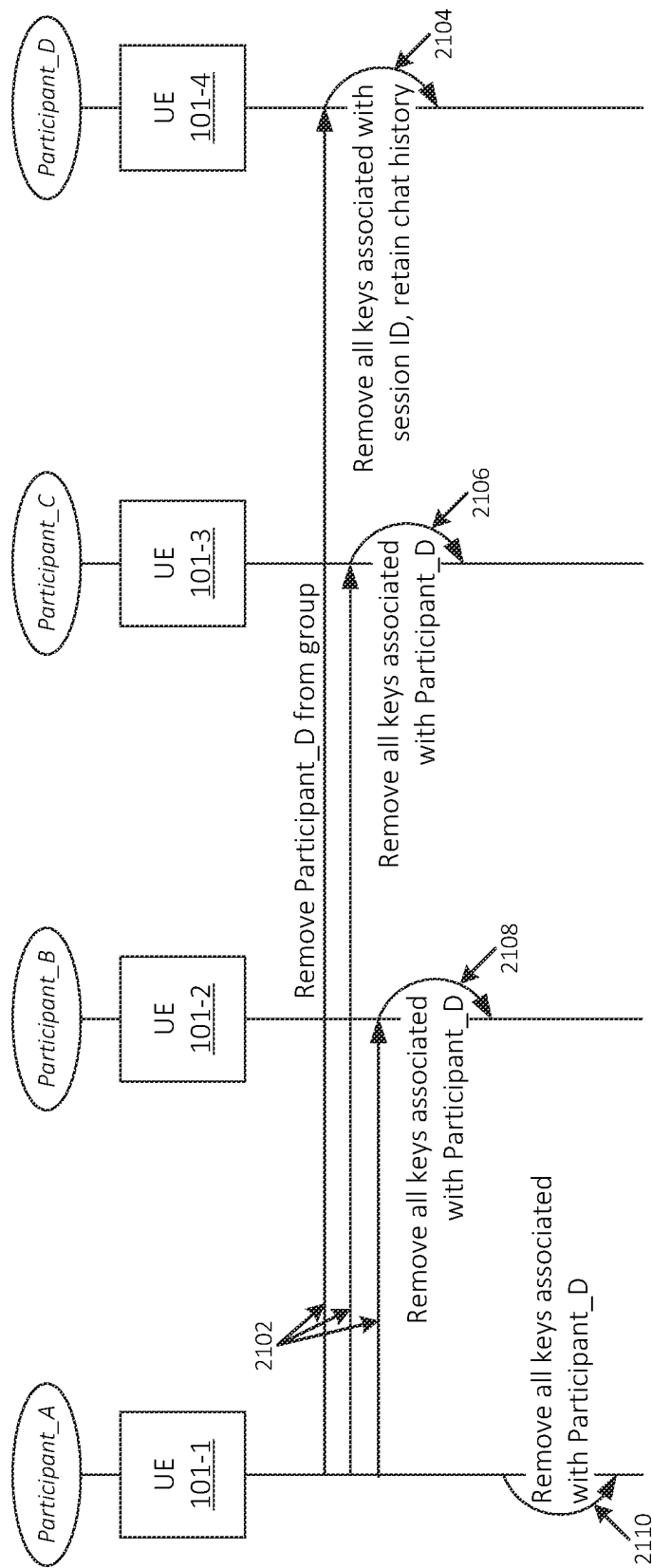


FIG. 21

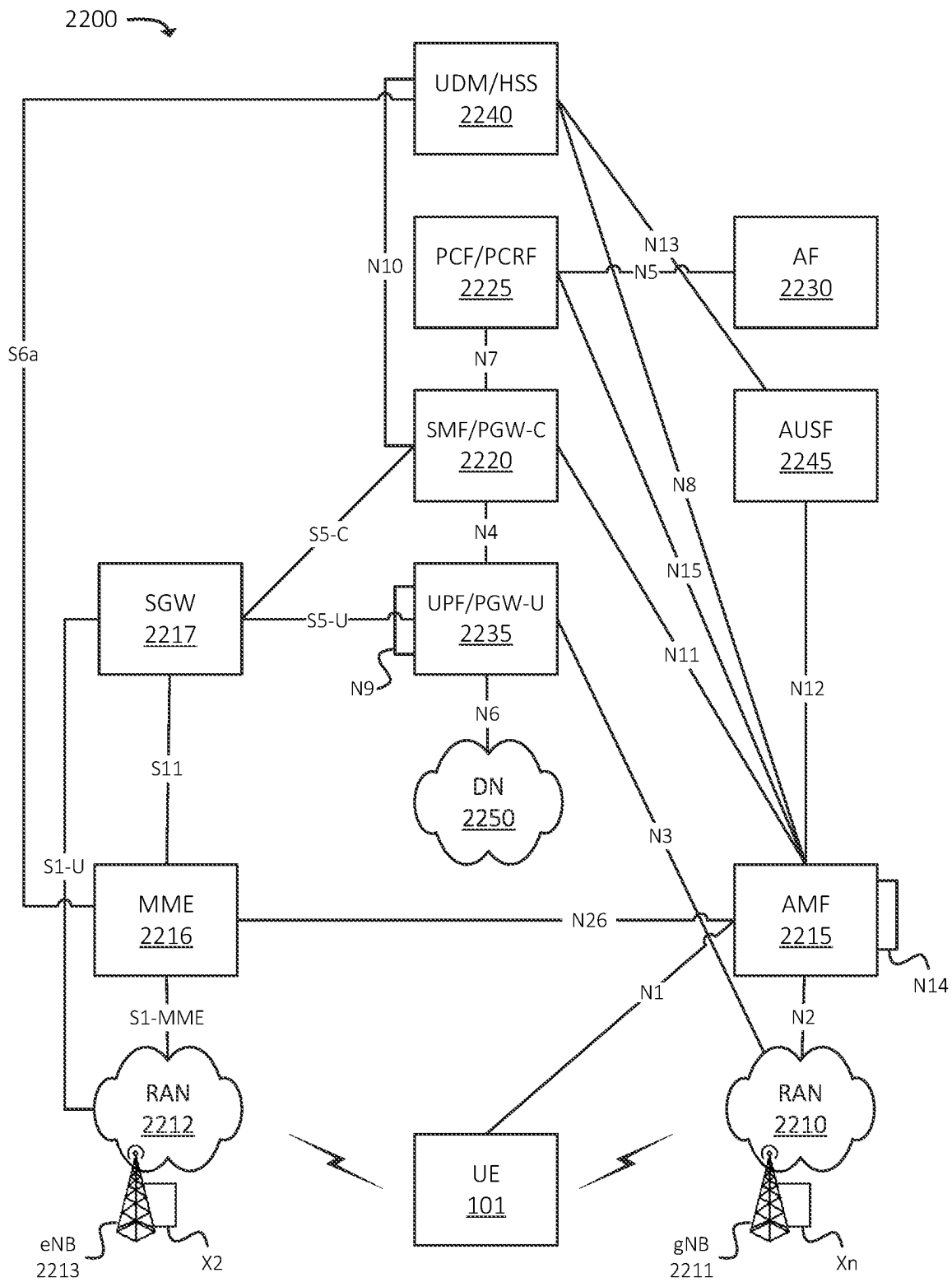


FIG. 22

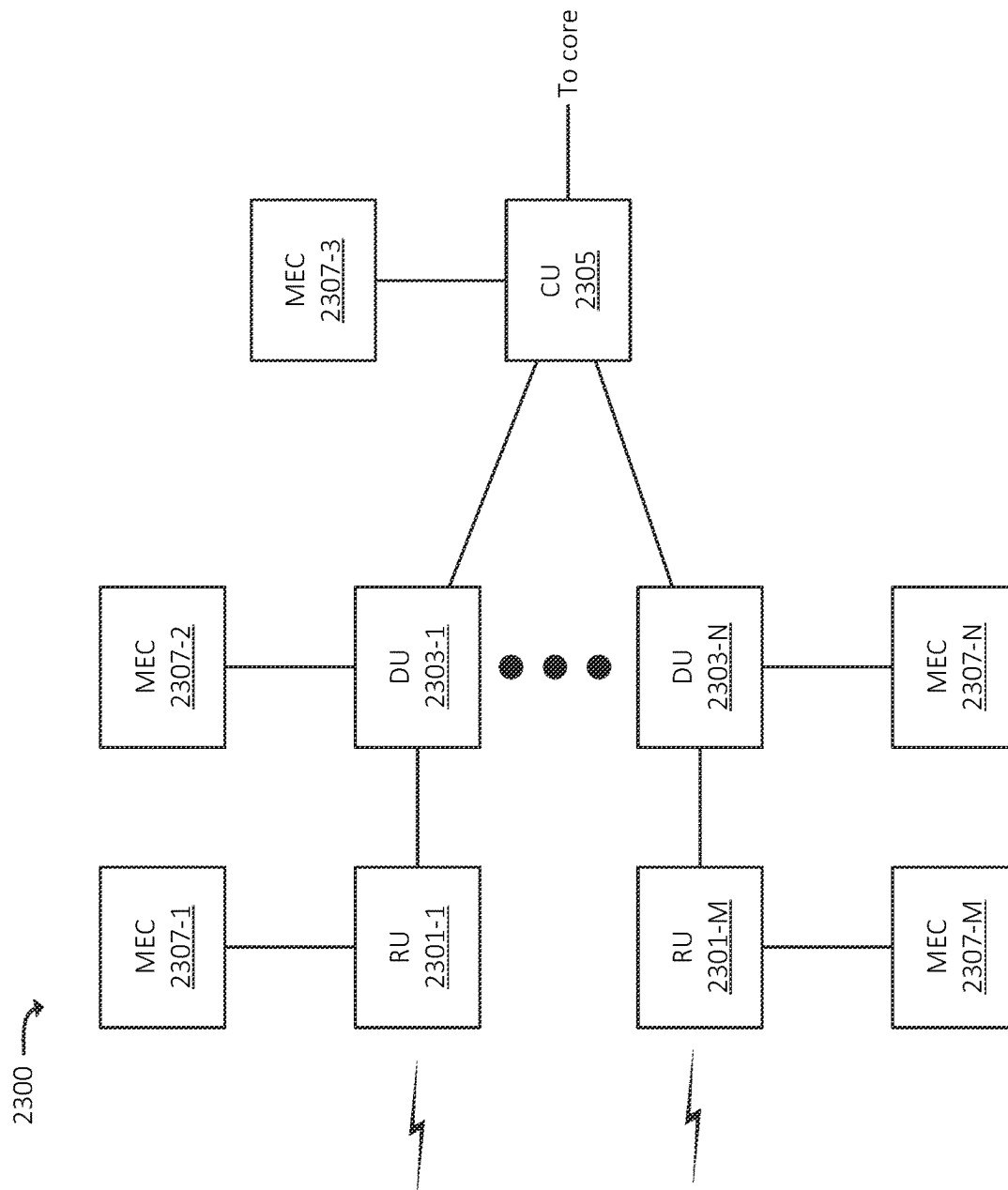
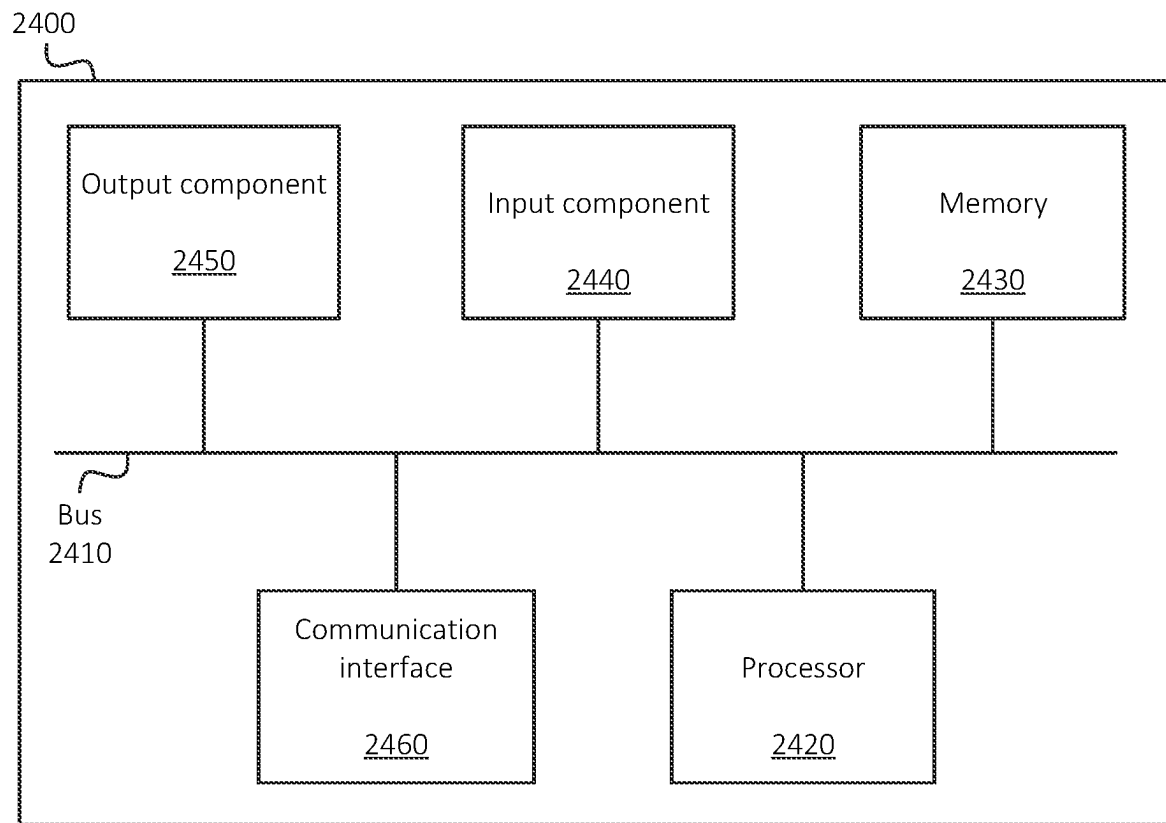


FIG. 23

**FIG. 24**

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SYSTEMS AND METHODS FOR GROUP MESSAGING USING BLOCKCHAIN-BASED SECURE KEY EXCHANGE

CROSS-REFERENCE TO RELATED APPLICATION

This Application is a Continuation-in-Part of U.S. patent application Ser. No. 17/321,378 filed on May 14, 2021, titled “SYSTEMS AND METHODS FOR BLOCKCHAIN-BASED SECURE KEY EXCHANGE,” the contents of which are herein incorporated by reference in their entirety.

BACKGROUND

Some encryption techniques, such as Public Key Infrastructure (“PKI”) techniques, may make use of public keys, which may be used to encrypt messages that may be decrypted using an associated private key, and/or may be used in other techniques. Public keys may be distributed or stored by key escrow systems, which may provide a single point of failure and/or an opportunity for a “man-in-the-middle” attack, via which public key may be obtained, modified, and/or otherwise compromised.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example overview of one or more embodiments described herein;

FIG. 2 illustrates an example of using a shared public key via a private blockchain system, in accordance with some embodiments, in a communication that utilizes double ratchet techniques to secure communications;

FIG. 3 illustrates an example of establishing a private blockchain system, in accordance with one or more embodiments;

FIG. 4 illustrates an example of using a shared public key via a private blockchain system, in accordance with some embodiments, in a communication that utilizes double ratchet techniques to secure communications;

FIGS. 5 and 6 illustrate example processes for using a shared public key via a private blockchain system, in accordance with some embodiments, in a communication that utilizes double ratchet techniques to secure communications;

FIG. 7 illustrates an example initiation of a group communication session, in accordance with some embodiments;

FIG. 8 illustrates an example sharing of public keys, along with associated key identifiers, to a blockchain system, in accordance with some embodiments;

FIG. 9 illustrates an example of obtaining and/or cross-validating public keys associated with active participants of a group communication session, in accordance with some embodiments;

FIG. 10 illustrates an example of outputting an encrypted message package to multiple participants of a group communication session, in accordance with some embodiments;

FIG. 11 illustrates an example of recipients of an encrypted message package obtaining and/or cross-validating a public key of a sender of the encrypted message from a blockchain system, in accordance with some embodiments;

FIGS. 12-16 illustrate further examples of blockchain-based secure group communications, in accordance with some embodiments;

FIGS. 17 and 18 illustrate an example automatic retrieval and/or cross-validation of public keys associated with par-

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ticipants of group communication session from a blockchain system, in accordance with some embodiments;

FIGS. 19 and 20 illustrate an example of adding a participant to an existing group communication session, in accordance with some embodiments;

FIG. 21 illustrates an example of removing a participant from a group communication session, in accordance with some embodiments;

FIG. 22 illustrates an example environment in which one or more embodiments, described herein, may be implemented;

FIG. 23 illustrates an example arrangement of a radio access network (“RAN”), in accordance with some embodiments; and

FIG. 24 illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

Embodiments described herein provide for the secure maintaining and providing of information, such as public keys used in PKI techniques or other techniques, using a secure distributed ledger (e.g., “blockchain”) system. For example, embodiments described herein may utilize a blockchain system in lieu of a key escrow system in the exchange and/or providing of public keys in a Diffie-Hellman key exchange technique or other type of technique in which public keys are provided from one entity to another. For example, as discussed herein, a first entity may generate an asymmetric key pair that includes a public key and a private key, and may provide the public key to a blockchain system for retrieval by one or more other entities. For example, the entities may be engaged in a secure messaging session, in which messages are encrypted and may be decrypted using one or more keys, including the public key. For example, the messaging session may be associated with multiple key derivation functions, in which a first key or set of keys (e.g., “root” keys) are used in the generation of one or more other keys (e.g., “send” keys and/or “receive” keys). A root key derivation function, for example, may be performed by first and second entities engaged in the communication session to generate respective send and/or receive keys for each entity. The entities may make use of different private keys in their respective root derivation functions, but may utilize one or more shared public keys for their respective root derivation functions. Examples of techniques that make use of multiple key derivation functions, including a root derivation function utilizing one or more shared public keys, include a “double ratchet” encryption technique, a “Signal Protocol,” and/or other types of techniques.

In some embodiments, the contents of the secured ledger system (e.g., blockchain) may be publicly available or accessible, but the information stored therein may have no meaning or use to an attacker or other malicious user, in the context of obtaining public keys associated with PKI techniques or other techniques. In some embodiments, a private blockchain may be used, in which only authorized entities are able to access the information stored in the private blockchain. Further, as records stored in a blockchain system are immutable, an attacker may be prevented from modifying public keys and thereby potentially having access to information encrypted based on such keys. For simplicity,

embodiments described herein are described in the context of the secured ledger system being a blockchain system. However, similar concepts may apply to another type of secured ledger system other than a blockchain system.

As shown in FIG. 1, a first entity, such as User Equipment (“UE”) 101-1, may generate (at 102) a set of keys. The set of keys may be, and/or may include, asymmetric key pair 103, which may include private key 105 and public key 107. Private key 105 may be used to decrypt information encrypted by public key 107, may be used in a double ratchet and/or Signal Protocol technique, and/or may be used for other suitable purposes. UE 101-1 may securely store private key 105, such as in a local storage device associated with UE 101-1, in a private key store, in a Universal Integrated Circuit Card (“UICC”), and/or in some other secure location.

As further shown, UE 101-1 may record (at 104) the public key to a blockchain system, such as private blockchain system 109. As noted above, private blockchain system 109 may be “private,” in that only authorized entities (e.g., authorized UEs, authorized users, etc.) may have access to the information stored in private blockchain system 109. The information stored in private blockchain system 109 may be organized as “blocks,” where one particular record corresponds to one particular block, and/or where one block includes multiple records. A block may also include a reference to one or more other blocks (e.g., a “previous” block and/or a “next” block), such that a group of blocks and their associated references may form a “chain” of blocks. Further, private blockchain system 109 may include a set of nodes 111 that form a consensus regarding the blocks of which the blockchain is comprised, thus preventing unauthorized changes to the information stored in the blockchain, as well as resiliency in case any particular node 111 is compromised or otherwise becomes unavailable. In some embodiments, nodes 111 may each be implemented by a respective UE 101, such as a mobile telephone, a workstation computer, a tablet, an Internet of Things (“IoT”) device, and/or other suitable type of device. While shown in the figure as being “external” to private blockchain system 109, in practice, UE 101-1 may be, and/or may implement, a particular node 111 associated with private blockchain system 109.

When recording the public key to private blockchain system 109, UE 101-1 may provide an identifier associated with UE 101-1 and/or other identifying information based on which the public key may be retrieved by an entity that may use the public key. For example, the record may include an Internet Protocol (“IP”) address associated with UE 101-1, a device name, a user name, a Session Initiation Protocol (“SIP”) address, and/or some other suitable information associated with UE 101-1.

Additionally, or alternatively, the record may include a session identifier, which may identify a communication session between UE 101-1 and another UE, such as UE 101-2. For example, UE 101-1 and UE 101-2 may be engaged in a secure messaging session, in which UE 101-1 and UE 101-2 exchange encrypted messages. The messaging session may be associated with a double ratchet technique, a Signal Protocol technique, and/or other type of technique in which UE 101-1 and UE 101-2 provide public keys to each other.

Briefly, for example, for each message sent between UE 101-1 and UE 101-2, a new public key may be generated (e.g., in an alternating fashion, where UE 101-1 generates a public key for a first message, UE 101-2 generates a public key for a second message, UE 101-1 generates a public key

for a third message, and so on). As described below, these alternating shared public keys may serve as a first “ratchet” in a double ratchet technique to ensure security of the encrypted communications. For example, as discussed below, the public keys may be used to generate symmetric send and/or receive keys, which may be used to encrypt and/or decrypt communications between UE 101-1 and UE 101-2 without transmitting the send and/or receive keys between each other.

Private blockchain system 109 is illustrated as including four nodes 111-1, 111-2, 111-3, and 111-4. In practice, private blockchain system 109 may include fewer nodes and/or additional nodes. Private blockchain system 109 may, in some embodiments, be associated with one or more suitable authentication mechanisms whereby only authorized devices or systems are able to participate as nodes 111 of private blockchain system 109. For example, as discussed herein, particular users or UEs 101 may be specified or automatically identified (e.g., based on an address book or contact list associated with a particular UE 101 that is authorized to access private blockchain system 109 and/or some other suitable technique). As such, unauthorized devices, such as UE 101-N, may not be able to access information stored in private blockchain system 109 and/or may not be able to participate as a node 111 of private blockchain system 109. Further, as the nodes 111 of private blockchain system 109 may only include trusted devices, the integrity of the information stored by private blockchain system 109 may be enhanced, as the likelihood of a compromise or attack of private blockchain system 109 may be reduced or eliminated. Further, the recording and/or accessing of private blockchain system 109 may be private with respect to external devices that do not have access to private blockchain system 109.

In some embodiments, separate private blockchain systems 109 may be established for discrete groups, such as messaging or conversation groups (e.g., where a first communication session includes a first set of participants and is associated with a first private blockchain system 109, and where a second communication session includes a second set of participants and is associated with a second private blockchain system 109). In some embodiments, a single private blockchain system 109 may be associated with multiple communication sessions, where keys associated with a given communication session may be protected by a set of smart contracts or other security mechanisms that are accessible only to participants in the communication session.

Once recorded to private blockchain system 109, public key 107 may be obtained (at 106) by UE 101-2 and/or another device or system to which the information stored in private blockchain system 109 is available. For example, as noted above, UE 101-2 may be a UE with which UE 101-1 is engaged in a communication session via which UE 101-1 and UE 101-2 exchange encrypted communications. UE 101-2 may identify a record stored in private blockchain system 109 that includes the identifier associated with UE 101-1, a communication session associated with UE 101-1 and UE 101-2, and/or some other suitable identifier. In some embodiments, UE 101-2 may be a node 111 of private blockchain system 109, and may maintain a full copy of the blockchain stored by private blockchain system 109. In some embodiments, private blockchain system 109 may implement or be communicatively coupled to a search system, which may provide records that match a query. For example, UE 101-2 may query private blockchain system 109 for the identifier associated with UE 101-1 and/or a

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communication session with UE 101-1, and private blockchain system 109 may return one or more blocks that include the identifier associated with UE 101-1 and/or the communication session. In some embodiments, private blockchain system 109 may return, and/or UE 101-2 may otherwise identify, the latest (e.g., most current) block matching the query. In this manner, in situations where UE 101-1 has provided multiple public keys to private blockchain system 109 over time, UE 101-2 may be able to retrieve and/or otherwise identify the newest public key. Identifying the newest public key may be useful in a double ratchet technique, a Signal Protocol technique, and/or some other technique in which an updated public key is used for each message between UE 101-1 and UE 101-2. In some embodiments, public keys may be provided with sequence numbers or other information, based on which public keys associated with communications received in a non-sequential manner (e.g., out of order) may be identified.

In some embodiments, the public key may be provided as part of a Diffie-Hellman exchange or some other suitable technique in which public keys associated with multiple UEs 101 are exchanged amongst each other. UE 101-2 may accordingly maintain and/or use (at 108) the received public key for a double ratchet technique, a Signal Protocol technique, and/or in some other technique that involves a Diffie-Hellman key exchange or other type of key exchange procedure.

FIG. 2 illustrates an example of using a shared public key (e.g., shared via private blockchain system 109, in accordance with some embodiments) in a communication technique that utilizes double ratchet techniques to secure communications between UE 101-1 and UE 101-2. As shown, UE 101-1 and UE 101-2 may each include a respective messaging client 201, which may send and/or receive encrypted communications. For example, messaging client 201-1, associated with UE 101-1, may encrypt communications based on techniques described herein, and may send the encrypted communications to messaging client 201-2 associated with UE 101-2. Messaging client 201-2 may receive encrypted communications from UE 101-1 (e.g., from messaging client 201-1), may decrypt the communications based on techniques described herein, and may present the decrypted communications (e.g., via a graphical user interface (“GUI”) or other suitable manner of presentation). Similarly, messaging client 201-2, associated with UE 101-2, may encrypt communications based on techniques described herein, and may send the encrypted communications to messaging client 201-1 associated with UE 101-1. Messaging client 201-1 may receive encrypted communications from UE 101-2 (e.g., from messaging client 201-2), may decrypt the communications based on techniques described herein, and may present the decrypted communications (e.g., via a GUI or other suitable manner of presentation).

The encryption and/or decryption by messaging client 201-1 and/or messaging client 201-2 may be performed using a symmetric key that is generated based on a double ratchet technique, a Signal Protocol technique, and/or other suitable technique. For example, as discussed herein, the symmetric key may be used by a sender (e.g., UE 101-1, in this example) to encrypt a communication, and may be used by a recipient (e.g., UE 101-2, in this example) to decrypt the communication. The symmetric keys may be generated by respective Send Key Derivation Function (“KDF”) 205 and/or Receive KDF 207 associated with the sender and/or recipient. The symmetric keys may be generated based on one or more root keys provided by Root KDF 203 to Send

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KDF 205 and/or Receive KDF 207. For example, Root KDF 203-1 may provide a root key output based on a public key and a private key to Send KDF 205-1, which may generate a send key (e.g., based on the root key output and one or more private keys) which may be used to encrypt communications to be sent by messaging client 201-1. The private key based on which the root key output is generated may be privately maintained by UE 101-1 (e.g., not sent to UE 101-2), while the public key may be shared with UE 101-2 via private blockchain system 109, in accordance with embodiments described herein. Root KDF 203-1 may provide a different root key output for each message sent and/or received by messaging client 201-1, based on a public key exchange (e.g., via a Diffie-Hellman key exchange or other suitable procedure) with UE 101-2, which may be performed each time a message is sent by UE 101-1 and/or UE 101-2.

For example, when UE 101-1 sends a message, Root KDF 203-1 may generate a root key output, and may further output a public key based on which the root key output was generated, to private blockchain system 109. As discussed above, the public key may be provided with a session identifier (e.g., to identify a communication session between messaging client 201-1 and messaging client 201-2), an identifier associated with UE 101-1, an identifier associated with UE 101-2, and/or other suitable information based on which UE 101-2 may identify that the public key recorded to private blockchain system 109 is associated with the communication session between UE 101-1 and UE 101-2.

UE 101-2 (e.g., Root KDF 203-2) may obtain the root public key from private blockchain system 109 based on the session identifier or other suitable identifier, and may generate a root key output based on the obtained public key and a private key (e.g., a different private key than the private key used by Root KDF 203-1 to encrypt the communication). As noted above, using private blockchain system 109 to exchange public keys may maintain the integrity of the keys, such as by preventing malicious actors from modifying or “spoofing” keys. Further, using private blockchain system 109 may aid in situations where a given UE 101 (e.g., UE 101-2, in this example) is “offline” or is otherwise unavailable to receive public keys from another UE 101 (e.g., UE 101-1, in this example) at the time that the other UE 101 outputs the public keys.

Root KDF 203-2 may provide the root key output to Receive KDF 207-2, which may generate a receive key based on the root key output and one or more private keys. In accordance with the double ratchet techniques, Signal Protocol techniques, etc., the receive key generated by Receive KDF 207-2 may be the same as the send key output generated by Send KDF 205-1. In this manner, these respective send and receive keys may be an identical symmetric key that may be used to encrypt and decrypt communications, such as the encrypted communication sent by messaging client 201-1 to messaging client 201-2 in this example.

While an example is provided here in the context of UE 101-1 encrypting a communication, sending the encrypted communication to UE 101-2, and UE 101-2 decrypting the communication, similar techniques may be performed (e.g., iteratively) for communications encrypted and sent by UE 101-2 to UE 101-1, as denoted by the dashed lines in FIG. 2. For example, Root KDF 203-2 may generate a root key pair, output a public key of the root key pair to private blockchain system 109 for retrieval by UE 101-1, generate a root key output based on the root key pair, and provide the root key output to Send KDF 205-2. Send KDF 205-2 may

generate a send key based on the root key output and one or more private keys, and messaging client 201-2 may encrypt a communication based on the send key. Further, Root KDF 203-1 may retrieve the public key from private blockchain system 109, generate a root key output based on the retrieved public key and one or more private keys, and provide the root key output to Receive KDF 207-1. Receive KDF 207-1 may generate a receive key based on the root key output and one or more private keys, and messaging client 201-2, as encrypted based on a matching send key.

As noted above, private blockchain system 109 may be a “private” blockchain, in that only approved UEs 101 (or other suitable devices or systems) may participate in private blockchain system 109 as nodes 111 of private blockchain system 109. In some embodiments, eligible participants may be identified based on an automated determination of contacts associated with one or more participant nodes 111 of private blockchain system 109, and/or some other suitable technique. For example, as shown in FIG. 3, UE 101-1 may establish (at 301) and/or join private blockchain system 109. For example, UE 101-1 may establish a new private blockchain system 109 by generating a “genesis” block, and/or may join an existing private blockchain system 109. In some embodiments, UE 101-1 may communicate with one or more devices or systems that facilitate the establishment of private blockchain system 109.

UE 101-1 may further automatically identify (at 303) candidate participants (e.g., nodes 111 for private blockchain system 109, devices or systems that are authorized to access information stored in private blockchain system 109, or the like). For example, UE 101-1 may identify contacts in an address book, call history, and/or other type of contact list associated with UE 101-1. Additionally, or alternatively, UE 101-1 and/or some other device or system may utilize artificial intelligence/machine learning (“AI/ML”) techniques or other suitable techniques to identify UEs 101 with which UE 101-1 has communicated.

UE 101-1 may output (at 305) an invitation to join (e.g., as a node 111 and/or as an authorized viewer of information stored in private blockchain system 109) to identified UEs 101-2, 101-3, and 101-4. In some embodiments, UE 101-1 (e.g., an application or application programming interface (“API”) associated with UE 101-1) may communicate with contacts in a contact list associated with UE 101-1 (e.g., where the contacts include UEs 101-2, 101-3, and 101-4 in this example) via one or more suitable messaging or other types of communication applications, and may identify particular ones of the contacts participate in a key escrow or other type of key exchange. UE 101-1 may output an invitation to private blockchain system 109 to the identified contacts via the messaging or other communication applications and/or some other suitable communication pathway. In some embodiments, the invitation may include identifying information associated with UE 101-1, such as an IP address associated with UE 101-1, a Mobile Directory Number (“MDN”) associated with UE 101-1, an International Mobile Station Equipment Identity (“IMEI”) associated with UE 101-1, an International Mobile Subscriber Identity (“IMSI”) associated with UE 101-1, and/or some other suitable identifier.

UEs 101-2, 101-3, and 101-4 may accordingly join (at 307, 309, and 311, respectively) private blockchain system 109. For example, such UEs 101-2, 101-3, and 101-4 may register as nodes 111 using information provided (at 305) by UE 101-1, and/or may access information stored in private

blockchain system 109 using such information. In this manner, private blockchain system 109 may be “private” inasmuch as UEs 101 which were not invited (at 305) to private blockchain system 109 may not access the information stored in private blockchain system 109.

FIG. 4 illustrates an example of the use of private blockchain system 109 to exchange public keys in the encryption and/or decryption of secure communications between UEs 101 participating in a secure communication session. In this example, UEs 101-1, 101-2, 101-3, and 101-4 may be nodes 111 of private blockchain system 109. Further, UEs 101-1 and 101-3 may be participants in a communication session (e.g., via respective messaging clients 201 associated with UEs 101-1 and 101-3). For example, UE 101-1 and UE 101-3 may establish (at 401) a secure communication session using any suitable session establishment technique or protocol. The communication session may be associated with a session identifier or other suitable mechanism by which UEs 101-1 and/or 101-3 may identify the communication session. For example, an initiator of the communication session may generate or otherwise determine an identifier for the communication session, and/or UEs 101-1 and 101-3 may otherwise negotiate or determine an identifier for the communication session.

UE 101-1 may further generate (at 403) an asymmetric key pair, which may be used as part of a double ratchet technique, a Signal Protocol technique, and/or some other encryption technique used to secure communications between UEs 101-1 and 101-3. For example, the asymmetric key pair generated (at 403) by UE 101-1 may include a public key used in the generation of a send key used to encrypt communications sent to UE 101-3. While not explicitly shown in FIG. 4, UE 101-3 may also generate one or more keys that may be used in the encryption and/or decryption of messages between UEs 101-1 and 101-3, in accordance with the double ratchet technique, Signal Protocol technique, and/or other suitable encryption technique.

UE 101-1 may further provide (at 405) the public key of the generated asymmetric key pair to private blockchain system 109. As noted above, UEs 101-1, 101-2, 101-3, and 101-4 may be nodes 111 of private blockchain system 109. In other examples, UE 101-3 (e.g., a participant in the secure communication session) may not be a node 111 of private blockchain system 109, and thus UE 101-1 may not directly provide (at 405) the public key to UE 101-3. When providing (at 405) the public key to private blockchain system 109, UE 101-1 may include the session identifier of the communication session between UEs 101-1 and 101-3, and/or some other suitable identifier based on which UE 101-3 may identify (at 407) the public key provided by UE 101-1.

For example, as discussed above, UE 101-3 may retrieve (at 407) the public key from private blockchain system 109 based on identifying a record in private blockchain system 109 that includes the session identifier or other suitable identifier associated with UEs 101-1 and/or 101-3. In situations where multiple blocks exist with the session identifier, UE 101-3 may identify a latest (e.g., newest) block. For example, in accordance with the double ratchet and/or Signal Protocol techniques, a new public key may be provided each time a message is sent and/or received by UE 101-1 and/or UE 101-3. As such, identifying a latest public key associated with the communication session between UEs 101-1 and 101-3 may facilitate the secure communications according to the double ratchet and/or Signal Protocol techniques.

UE 101-1 may further encrypt (at 409) a message for UE 101-3 using a symmetric key (e.g., a send key, as similarly

discussed above) that UE 101-1 generates based on the public key, one or more private keys, and/or one or more KDFs (e.g., Send KDF 205). UE 101-1 may proceed to output (at 411) the encrypted message to UE 101-3. UE 101-3 may decrypt (at 413) the message using a symmetric key (e.g., the same symmetric key as UE 101-1 used to encrypt (at 409) the message), which may be generated by one or more KDFs (e.g., Receive KDF 207) based on the retrieved (at 407) public key.

While FIG. 4 shows an example of one message being sent from UE 101-1 to UE 101-3, similar techniques may be employed when UE 101-1 sends additional messages to UE 101-3, and/or when UE 101-3 sends messages to UE 101-1. For example, UE 101-3 may generate a subsequent asymmetric key pair that is derived from and/or is otherwise based on the retrieved (at 407) public key, and may output a public key of the subsequent asymmetric key pair to private blockchain system 109. UE 101-3 may utilize this subsequently generated public key to generate a send key, and UE 101-1 may utilize this subsequently generated public key, retrieved from private blockchain system 109, to generate a receive key, as described above.

FIGS. 5 and 6 illustrate example processes 500 and 600, respectively, for utilizing blockchain techniques to share and/or exchange public keys. In the examples provided herein, the key exchange may be part of a Diffie-Hellman key exchange, a key exchange portion of a Signal Protocol technique, a key exchange portion of a double ratchet technique, and/or some other suitable secure exchange of public keys. In some embodiments, some or all of process 500 and/or 600 may be performed by one or more UEs 101 (e.g., UEs 101 engaged in a communication session with each other and/or one or more other UEs 101). In some embodiments, one or more other devices may perform some or all of processes 500 and/or 600 in concert with, and/or in lieu of, UE 101.

As shown, process 500 may include establishing (at 502) a communication session with one or more UEs 101. The communication session may be associated with a session identifier or other identifier based on which participant UEs 101 of the communication session may identify messages exchanged with one another and/or public keys, associated with the communication session, recorded to private blockchain system 109 in accordance with embodiments described herein.

Process 500 may further include generating (at 504) a set of root keys, which may include an asymmetric key pair. For example, as discussed above, the root keys may include a private key which may be maintained securely by UE 101, and a public key which may ultimately be shared with one or more other UEs 101 (e.g., another participant in the communication session).

Process 500 may additionally include recording (at 506) the public root key (e.g., the public key of the asymmetric key pair generated at 504) to private blockchain system 109. For example, UE 101 may output the public key to private blockchain system 109, and private blockchain system 109 may propagate the information to one or more nodes 111 of private blockchain system 109. Nodes 111 may form a consensus regarding the addition of the provided public key to one or more records of private blockchain system 109, and may maintain an immutable record of the provided public key. In some embodiments, the record may include a timestamp, a block identifier, and/or other mechanism by which a recency or age of the record may be determined. As similarly described above, the record may further include a

communication session identifier, an identifier of UE 101 from which the information was received, and/or other suitable information.

Process 500 may also include utilizing (at 508) the root keys to generate a send key. For example, as discussed above, Root KDF 203 may utilize the root keys (e.g., including the root public key, as well as the root private key and/or one or more other keys) to generate a root key output, and may provide the root key output to another KDF, such as Send KDF 205. Send KDF 205 may generate a send key based on the root key output from Root KDF 203 and/or based on one or more other keys or functions. As discussed above, the generated send key may be a symmetric key that may be identical to a receive key generated by a recipient of the public key, based on a Signal Protocol technique, a double ratchet technique, etc.

Process 500 may further include encrypting (at 510) a communication using the send key. Process 500 may additionally include outputting (at 512) the encrypted communication via the communication session, such as to a UE 101 with which the communication session was established (at 502).

As shown in FIG. 6, process 600 may include establishing (at 602) a communication session. For example, as similarly discussed above, UE 101 may establish a communication session with one or more other UEs 101. The communication session may be associated with a suitable identifier, based on which the UEs 101 participating in the communication session may identify communications associated with the communication session, including encrypted messages, public keys, and/or other information.

Process 600 may further include receiving (at 604) an encrypted communication via the communication session. For example, the communication may have been encrypted using one or more KDFs, where the input to the one or more KDFs include the public root key and one or more other keys (e.g., one or more private keys, one or more KDF outputs, etc.).

Process 600 may further include retrieving (at 606) a public root key from private blockchain system 109. In some embodiments, private blockchain system 109 and/or one or more devices or systems communicatively coupled to private blockchain system 109 may “push” the public root key to private blockchain system 109. For example, the one or more devices or systems may identify a particular block that includes an identifier of UE 101 (e.g., where the particular block includes the identifier of UE 101 and a public root key), may identify a particular block that includes an identifier of the communication session (e.g., where the particular block includes the identifier of the communication session and a public root key), etc. Additionally, or alternatively, UE 101 may “pull” the information from private blockchain system 109 based on a suitable identifier (e.g., an identifier of UE 101, an identifier of the communication session, etc.).

Process 600 may additionally include utilizing (at 608) the public root key to generate a receive key. For example, as discussed above, UE 101 may utilize one or more KDFs, such as Root KDF 203 and/or Receive KDF 207, to generate a receive key. The receive key may be a symmetric key with respect to a key used to encrypt a communication received by UE 101 via the communication session. For example, as discussed above, the receive key may be identical to a send key used by a sender of the communication, where the send key was also generated based on one or more KDFs and at least the public root key. For example, the sender and UE 101 may perform double ratchet techniques, Signal Protocol

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techniques, or the like, to respectively encrypt and decrypt the communication, without transmitting or sharing the symmetric key itself.

Process 600 may further include decrypting (at 610) the communication, received via the communication session, using the generated receive key. For example, UE 101 may apply one or more functions to decrypt the receive message using the receive key, which, as discussed above, may be the same as a send key used to encrypt the communication.

Process 600 may additionally include generating (at 612) a new set of root keys based on the previously retrieved public root key and one or more other root keys. For example, once UE 101 generates the receive key, the public root key (retrieved at 606) may be discarded and/or not used for sending messages from UE 101. In some situations, UE 101 may maintain the public root key and apply techniques described above to decrypt subsequent communications from the sender in situations where the sender sends multiple communications in a row without any intervening messages from UE 101. The new set of root keys may be generated based on the receive key (generated at 608), the public root key (retrieved at 606), and/or one or more other suitable keys. UE 101 may then proceed to output the newly generated public root key to private blockchain system 109, as similarly described above with respect to operation 506 of process 500, may generate (e.g., similar to operation 508) a new send key based on the newly generated public root key, etc.

Embodiments described above are presented in the context of communication sessions between two UEs 101. Embodiments described herein may further provide for secure blockchain-based key exchange mechanisms for group communication sessions that include more than two participants (e.g., between three or more UEs 101). For example, techniques described below provide for a blockchain-based key exchange such that group messages may be encrypted such that only participants in the group are able to access (e.g., decrypt) the messages. Further, as discussed below, situations may arise in which a participant in a communication session receives an encrypted message from another participant, but does not possess one or more keys needed to decrypt the message (e.g., a public key associated with the sender), and/or has not received confirmation from the blockchain that the participant is in possession of such keys. Techniques described below provide for the recipient to securely obtain the one or more keys from blockchain system 109 in order to decrypt the message. Further, embodiments described below provide for the adding and removing of participants from a secure group communication session.

FIG. 7 illustrates an example establishment of a group communication session that uses blockchain-based secure key exchange techniques. In this example, assume that the group communication session is between UEs 101-1, 101-2, and 101-3 (also referred to as "Participant_A," "Participant_B," and "Participant_C"). As shown, UE 101-1 may initiate (at 702) a group communication session (also referred to as a "group chat"), such as by sending (at 702) one or more messages to UEs 101-2 and 101-3. For example, a messaging application (e.g., a particular messaging client 201) of UE 101-1 may communicate with the same or a different messaging application (e.g., respective messaging clients 201) of UEs 101-2 and 101-3. The message(s) sent (at 702) by UE 101-1 may include one or more Rich Communication Suite ("RCS") messages or messages of another type or protocol. In some embodiments, the message(s) may include a session identifier associated with the group com-

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munication session, may include identifiers associated with UEs 101-1, 101-2, and/or 101-3 (e.g., IP addresses, SIP addresses, MDNs, etc.), and/or one or more other suitable identifiers. In some embodiments, the message(s) may include one or more instructions, indications, requests, etc. to implement blockchain-based key exchange for the communication session. In this example, assume that UEs 101-1, 101-2, and 101-3 support and/or have accepted the instruction, request, etc. to utilize blockchain-based key exchange.

UEs 101-1, 101-2, and 101-3 may further select (at 704) passive participants, or nodes 111, for blockchain system 109 to securely maintain keys associated with the group communication session. For example, as similarly discussed above, UEs 101-1, 101-2, and 101-3 may automatically select contacts from respective address books, contacts lists, etc. In this example, UE 101-1 may select UE 101-4 as a node 111 for blockchain system 109, UE 101-2 may select UE 101-5 as a node 111 for blockchain system 109, and UE 101-3 may select UE 101-6 as a node 111 for blockchain system 109.

UEs 101-2 and 101-3 may indicate (at 706) their respective selections of nodes 111 to UE 101-1. For example, UE 101-2 may provide an identifier (e.g., MDN, IP address, etc.) of UE 101-5, and UE 101-3 may provide an identifier of UE 101-6. In some embodiments, UE 101-1 may have also provided (at 702) a public key to UEs 101-2 and 101-3. In such embodiments, UEs 101-2 and 101-3 may each encrypt the messages sent (at 706) to UE 101-1, including the identifiers of UEs 101-5 and 101-6.

UE 101-1 may generate (at 708) a full list of blockchain participants, including active participant UEs 101-1, 101-2, and 101-3, as well as passive participant UEs 101-4, 101-5, and 101-6. UE 101-1 may provide (at 710) the full list of blockchain participants to UEs 101-2 and 101-3, thus providing for UEs 101-2 and 101-3 to be aware of, and therefore able to communicate with, blockchain system 109 once established. In some embodiments, UE 101-1 may encrypt the message (sent at 710) with a private key, which may be associated with the public key provided at 702. For example, the public and private keys may be associated with the same asymmetric key pair. UEs 101-2 and 101-3 may be able to use the public key (received at 702) of UE 101-1 to decrypt the provided list of blockchain participants. UE 101-1 may proceed to establish (at 712) blockchain system 109, which may include generating a genesis block, providing the genesis block to nodes 111 of blockchain system 109 (e.g., UEs 101-2 through 101-6), and/or other operations to establish blockchain system 109. In some embodiments, the genesis block or other data included in or associated with blockchain system 109 may include or may be based on the session identifier associated with the group communication session. In this manner, different blockchain systems 109 may be associated with different communication sessions and may have different session identifiers.

As shown in FIGS. 8-18, UEs 101-1, 101-2, and 101-3 may utilize blockchain-based key exchange techniques to securely communicate via the established group communication session. As shown in FIG. 8, UE 101-1 may generate asymmetric key pair 103-1, including private key 105-1 and public key 107-1. In embodiments where UE 101-1 generated a key pair and provided a public key (e.g., at 702) to UEs 101-1 and 101-2, asymmetric key pair 103-1 (shown in FIG. 8) may be a different key pair. UE 101-1 may further record public key 107-1 to blockchain system 109. Similarly, UE 101-2 may generate asymmetric key pair 103-2 (which may include private key 105-2 and public key 107-2) and may record public key 107-2 to blockchain system 109.

Additionally, UE 101-3 may generate asymmetric key pair 103-3 (which may include private key 105-3 and public key 107-3) and may record public key 107-3 to blockchain system 109.

In some embodiments, when recording keys (e.g., public keys 107) to blockchain system 109, respective UEs 101 may provide a key identifier. For example, UE 101-1 may provide key identifier 801-1 with public key 107-1, UE 101-2 may provide key identifier 801-2 with public key 107-2, and UE 101-3 may provide key identifier 801-3 with public key 107-3. In some embodiments, key identifiers 801 may be unique to each respective public key 107, such that particular public keys 107 may be retrieved from blockchain system 109 based on associated key identifiers 801. For example, in situations where a double ratchet encryption technique and/or other type of protocol in which keys are iteratively generated and used to encrypt messages associated with the group communication session, up-to-date public keys 107 may be retrieved from blockchain system 109 based on respective key identifiers 801, as discussed below.

In some embodiments, key identifiers 801 may include, and/or may be based on, identifiers of respective UEs 101 with which public keys 107 are associated. For example, key identifier 801-1 may include, may be derived from, and/or may otherwise be based on an identifier (e.g., MDN, IP address, etc.) associated with UE 101-1. Additionally, or alternatively, key identifier 801-1 may include, may be derived from, and/or may otherwise be based on a session identifier associated with the group communication session between UEs 101-1, 101-2, and 101-3. In some embodiments, key identifier 801-1 may be independent of, and/or may not be based on, a respective UE identifier and/or session identifier. In some embodiments, public keys 107 may be provided to blockchain system 109 with an indication of a respective UE identifier and/or session identifier (e.g., independent of key identifier 801).

As shown in FIG. 9, UE 101-1 may generate message 901, which may be associated with the group communication session between UEs 101-1, 101-2, and 101-3. For example, a user of UE 101-1 may have provided input such as text, audio, an image, etc. to be sent to UEs 101-2 and 101-3 associated with the group communication session. Such input may be provided, for example, via a suitable messaging application executing at UE 101-1. Based on receiving the input specifying outgoing message 901, UE 101-1 may determine that UE 101-1 does not have possession of public keys associated with the other active participants of the group communication session, namely public keys 107-2 and 107-3 associated with UEs 101-2 and 101-3. Additionally, or alternatively, UE 101-1 may determine that UE 101-1 is in possession of public keys 107-2 and 107-3 (e.g., may have received such public keys 107 by virtue of being a node 111 of blockchain system 109, as discussed above with respect to FIG. 8), but may determine that such public keys 107 have not been confirmed by blockchain system 109.

Accordingly, as shown, UE 101-1 may obtain public keys 107-2 and 107-3 from blockchain system 109. As discussed above, for example, public keys 107-2 and 107-3 may have previously been recorded to blockchain system 109 by UEs 101-2 and 101-3. In some embodiments, UE 101-1 may identify public keys 107-2 and 107-3 by identifying a session identifier recorded with keys 107-2 and 107-3 and/or UE identifiers of UE 101-2 and 101-3 recorded with public keys 107-2 and 107-3, as discussed above. In some embodiments, UE 101-1 may identify public keys 107-2 and 107-3

based on key identifiers 801-2 and 801-3 associated with public keys 107-2 and 107-3. For example, based on detecting that UE 101-1 does not have possession of such keys, and/or based on detecting that UE 101-1 has received input specifying outgoing message 901, UE 101-1 may output requests to UEs 101-2 and 101-3 for key identifiers 801-2 and 801-3 associated with public keys 107-2 and 107-3 of UEs 101-2 and 101-3. In such embodiments, UEs 101-2 and 101-3 may provide key identifiers 801-2 and 801-3, based on which UE 101-1 may search blockchain system 109 and/or otherwise identify corresponding public keys 107-2 and 107-3.

In some embodiments, when obtaining keys (e.g., public keys 107-2 and 107-3, in this example) from blockchain system 109, a particular UE 101 obtaining such keys (e.g., UE 101-1, in this example) may perform one or more cross-validation techniques in order to verify that the keys are accurate and/or have not been compromised in some way (e.g., a “man-in-the-middle” attack or some other type of malicious action). For example, in this example, UE 101-1 may compute a hash of one or more blocks of blockchain system 109 from which public keys 107-2 and/or 107-3 were obtained, and/or may otherwise identify or generate information based on information recorded to blockchain system 109 that corresponds to public keys 107-2 and 107-3.

UE 101-1 may further communicate with one or more other nodes 111 of blockchain system 109 (e.g., UEs 101-2, 101-3, 101-4, 101-5, and/or 101-6) in order to verify the information (e.g., the hash of the one or more blocks from which public keys 107-2 and/or 107-3 were obtained). When quorum is reached (e.g., UE 101-1 has received at least a threshold quantity of confirmations from other nodes 111) with respect to the hash or other suitable information, UE 101-1 may identify that the obtained public keys 107-2 and 107-3 are valid, accurate, etc., thus enhancing the security of the group communication session. Similar concepts may apply in situations where UE 101-1 already has possession of public keys 107-2 and 107-3 (e.g., in situations where UE 101-1 is a node 111 of blockchain system 109 and has therefore received public keys 107-2 and 107-3 when UEs 101-2 and 101-3 provided such public keys 107 to blockchain system 109). For example, UE 101-1 may identify a local copy of blockchain system 109 and may obtain public keys 107-2 and 107-3 from the local copy of blockchain system 109, and may perform a cross-validation process (e.g., may communicate with other nodes 111 of blockchain system 109) to verify that public keys 107-2 and 107-3 from the local copy of blockchain system 109 is accurate, valid, etc.

In other figures that discuss the obtaining of information (e.g., respective public keys 107) from blockchain system 109, a given UE 101 obtaining such information may perform similar operations in order to verify the authenticity, accuracy, etc. of the obtained information. Similarly, in other examples that discuss whether a given UE 101 is in possession of particular information (e.g., respective public keys 107), such examples may also refer to whether the given UE 101 has received confirmations from other nodes 111 (e.g., at least a threshold quantity of nodes 111, such that a quorum is reached) that such information is valid, even in instances where the given UE 101 already has possession of the information. That is, the given UE 101 may store a copy of one or more public keys 107 (e.g., within a local copy of blockchain system 109), but may not be considered in “possession” of such public keys 107, in accordance with some embodiments, if UE 101 has not received confirmation, validation, etc. from at least a threshold quantity of

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nodes 111 of blockchain system 109 that such copies of public keys 107 are valid, accurate, etc.

As shown in FIG. 10, after retrieving public keys 107-2 and 107-3 from blockchain system 109 (and/or after receiving confirmation from at least a threshold quantity of nodes 111 of blockchain system 109 that such public keys 107-2 and 107-3 are valid), UE 101-1 may have possession of validated copies of public keys 107-2 and 107-3 (also referred to in the figures as “Public key B” and “Public key C”), as well as its own private key 105-1 (also referred to in the figures as “Private key A”) and public key 107-1. As further shown, at this point, UEs 101-2 and 101-3 may have possession of their own respective private keys 105 and public keys 107, but may not yet have possession of private keys 105 or public keys 107 associated with other UEs 101 (and/or may not have confirmation that locally stored copies of private keys 105 or public keys 107, stored by UEs 101-2 and 101-3 in respective local copies of blockchain system 109, are valid).

UE 101-1 may generate encrypted message package 1001 based on message 901, and send encrypted message package 1001 to UEs 101-2 and 101-3. As shown, encrypted message package 1001 may include first portion 1003-1 and second portion 1003-2, as well as key identifier 801-1, associated with public key 107-1. Portion 1003-1 may include a first encrypted version of message 901, while portion 1003-2 may include a second encrypted version of message 901. The first encrypted version of message 901 (e.g., as included in portion 1003-1) may, in some embodiments, be encrypted based on private key 105-1 (e.g., Private key A) and public key 107-2 (e.g., Public key B). For example, the first encrypted version of message 901 may be encrypted based on a symmetric key derived from private key 105-1 and public key 107-2. Similarly, the second encrypted version of message 901 (e.g., as included in portion 1003-2) may be encrypted based on private key 105-1 and public key 107-3 (e.g., a symmetric key derived from Private key A and Public key C).

In some embodiments, portion 1003-1 may include an identifier of UE 101-2 and/or key identifier 801-2 associated with public key 107-2, such that UE 101-2 is able to identify that UE 101-2 is the recipient of portion 1003-1 and/or that public key 107-2 has been used in the encryption of message 901 included in portion 1003-1. Similarly, portion 1003-2 may include an identifier of UE 101-3 and/or key identifier 801-3, such that UE 101-3 is able to identify that UE 101-3 is the recipient of portion 1003-2.

As shown in FIG. 11, UEs 101-2 and 101-3 may retrieve public key 107-1 (e.g., Public key A) from blockchain system 109, based on receiving encrypted message package 1001. For example, UE 101-2 may identify that portion 1003-1 includes an encrypted message from UE 101-1 associated with the group communication session, and UE 101-3 may identify that portion 1003-2 includes an encrypted message from UE 101-1 associated with the group communication session. For example, as discussed above, encrypted message package 1001 and/or respective portions 1003 may include a session identifier associated with the group communication session, key identifier 801-1, and/or identifiers of UEs 101-2 and/or 101-3. As discussed above, retrieving public key 107-1 from blockchain system 109 may further include cross-validating the retrieved public key 107-1 with at least a threshold quantity of nodes 111 of blockchain system 109 (e.g., including some or all of UEs 101-1 through 101-6).

UEs 101-2 and 101-3 may accordingly use key identifier 801-1, a session identifier, and/or other suitable information

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to identify public key 107-1, as previously recorded to blockchain system 109. Once UEs 101-2 and 101-3 have obtained public key 107-1, UE 101-2 may be in possession of public key 107-1 as well as its own private key 105-2 and public key 107-2, and UE 101-3 may also be in possession of public key 107-1 as well as its own private key 105-3 and public key 107-3. UEs 101-2 and 101-3 may accordingly decrypt their own respective portions 1003-1 and 1003-2 in order to extract original message 901. For example, as similarly discussed above, UE 101-2 may derive a symmetric key based on private key 105-2 (e.g., Private key B) and public key 107-1 (e.g., Public key A), and may use the symmetric key to decrypt portion 1003-1 in order to extract message 901. Similarly, UE 101-3 may derive a symmetric key based on private key 105-3 (e.g., Private key C) and public key 107-1 (e.g., Public key A), and may use the symmetric key to decrypt portion 1003-2 in order to extract message 901.

As shown in FIG. 12, UE 101-3 may receive input specifying outgoing message 1201, associated with the same group communication session. Based on operations described above, UE 101-1 may be in possession of keys 105-1, 107-1, 107-2, and 107-3; UE 101-2 may be in possession of keys 105-2, 107-1, and 107-2; and UE 101-2 may be in possession of keys 105-3, 107-1, and 107-3. Based on receiving the input specifying message 1201, UE 101-3 may identify that UE 101-3 is not in possession of a public key associated with UE 101-2. Accordingly, as similarly discussed above, UE 101-3 may obtain public key 107-2, associated with UE 101-2, from blockchain system 109. For example, UE 101-3 may identify public key 107-2, as recorded to blockchain system 109, based on a session identifier recorded with public key 107-2 and/or a UE identifier of UE 101-2 recorded with public key 107-2. Additionally, or alternatively, as discussed above, UE 101-3 may communicate with UE 101-2 in order to obtain a respective key identifier 801-2 associated with public key 107-2, and may obtain public key 107-2 from blockchain system 109 based on key identifier 801-2.

In some embodiments, UE 101-3 may attempt to obtain a respective public key 107 associated with each group participant based on receiving the input specifying message 1201. That is, in some embodiments, each time a respective UE 101 receives input specifying an outgoing message, such UE 101 may attempt to obtain public keys 107 associated with other group participants, in implementations where new public keys 107 (e.g., associated with new key pairs 103) are generated by UEs 101. For example, as discussed above, such implementations may include a double ratchet technique, in which updated key pairs 103 are generated for different outgoing messages. In the examples shown in the figures, it is assumed that the same respective key pairs 103 are used by UEs 101, for the sake of simplicity.

As shown in FIG. 13, UE 101-3 may output encrypted message package 1301 to UEs 101-1 and 101-2. Encrypted message package 1301 may include, for example, encrypted portions 1303-1 and 1303-2, where portion 1303-1 includes a first encrypted version of message 1201 and portion 1303-2 includes a second encrypted version of message 1201. For example, the first encrypted version of message 1201 may be encrypted based on a symmetric key associated with UEs 101-1 and 101-3 (e.g., Public key A and Private key C), and the second encrypted version of message 1201 may be encrypted based on a symmetric key associated with UEs 101-2 and 101-3 (e.g., Public key B and Private key C).

As shown in FIG. 14, based on receiving encrypted message package 1301, UE 101-1 may decrypt portion

1303-1 using previously obtained public key **107-3** (e.g., as obtained when outputting a group message, as shown in FIG. 9). For example, UE **101-1** may derive a symmetric key based on private key **105-1** (e.g., Private key A) and public key **107-3** (e.g., Public key C).

Additionally, or alternatively, in some embodiments, UE **101-3** may have generated a new public key **107** (e.g., associated with a new private key **105**), and recorded the new public key **107** to blockchain system **109**. In such embodiments, UE **101-1** may request, and/or UE **101-3** may otherwise provide, a key identifier **801** associated with the new public key **107** to UE **101-1**, such that UE **101-1** is able to obtain the new public key **107** from blockchain system **109**. As further shown, UE **101-2** may obtain public key **107-3** from blockchain system **109** based on receiving encrypted message package **1301** (e.g., based on identifying portion **1303-2**). Once obtained, UE **101-2** may use public key **107-3** (e.g., along with private key **105-2**) to decrypt portion **1303-2** in order to extract message **1201**, as sent by UE **101-3**.

At this point, UEs **101-1**, **101-2**, and **101-3** may all be in possession of each other's public keys **107**, as well as their own respective private keys **105**. Thus, when sending and receiving messages via the group communication session, UEs **101-1**, **101-2**, and **101-3** may not need to access blockchain system **109** any more. Thus, even in situations where other nodes **111** of blockchain system **109** go offline or become unreachable, the security and decentralization provided by techniques described above may be maintained. For example, as shown in FIG. 15, UE **101-2** may generate encrypted message package **1501**, including portions **1503-1** and **1503-2**, which may each include a respective encrypted version of outgoing message **1551**. UE **101-2** may, in some embodiments, generate encrypted message package **1501** without attempting to access blockchain system **109** (e.g., communicating with nodes **111** of blockchain system **109**), as UE **101-2** is already in possession of the respective public keys **107** needed to encrypt message **1551** in accordance with example embodiments described above. Similarly, as shown in FIG. 16, UEs **101-1** and **101-3** may receive encrypted message package **1501** and decrypt respective portions **1503-1** and **1503-2** using previously obtained public key **107-2** (e.g., Public key B) and their respective private keys **105-1** and **105-3**, in order to extract message **1551**.

In some embodiments, participants in a group communication session may retrieve and/or cross-validate public keys **107** of other participants prior to receiving encrypted messages via the group communication session. For example, as shown in FIG. 17, when initiating a group communication session, UE **101-1** may provide session identifier **1701** to other participants of the group communication session (e.g., UEs **101-2** and **101-3**). As similarly discussed above, UEs **101-1**, **101-2**, and **101-3** may generate respective asymmetric key pairs **103**, including a respective private key **105** and public key **107**. UEs **101-1**, **101-2**, and **101-3** may record respective public keys **107** to blockchain system **109**, along with session identifier **1701**. In some embodiments, session identifier **1701** may be provided with public keys **107** along with other suitable information, such as key identifiers **801**, UE identifiers, etc.

As shown in FIG. 18, UEs **101-1**, **101-2**, and **101-3** may automatically retrieve and/or cross-validate public keys **107** of the other group communication session participants (e.g., independently of receiving any encrypted messages or encrypted message packages via the group communication session). For example, UEs **101-1**, **101-2**, and **101-3** may implement and/or communicate with a "listener" or

"explorer" of blockchain system **109**, based on which UEs **101-1**, **101-2**, and/or **101-3** may identify that public keys **107** of other group communication session participants have been recorded to blockchain system **109** (e.g., as discussed above with respect to FIG. 17). Whenever such public keys **107** are recorded to blockchain system **109** (e.g., along with session identifier **1701** associated with the group communication session), UEs **101-1**, **101-2**, and/or **101-3** may automatically retrieve such public keys **107**. Additionally, or alternatively, UEs **101-1**, **101-2**, and/or **101-3** may perform a cross-validation of public keys **107**, including communicating with other nodes **111** of blockchain system **109** (e.g., at least a threshold quantity of UEs **101** from the set of UEs **101-1** through **101-6** that serve as nodes **111** of blockchain system **109**), to verify that the retrieved public keys **107** are valid.

In some embodiments, participants may be added or removed from a group communication session secured by blockchain-based key exchange techniques. For example, as shown in FIG. 19, UE **101-1** may add (at **1902**) a participant to an existing group communication session. For example, UE **101-1** may add UE **101-4** to a group communication session that already includes UEs **101-1**, **101-2**, and **101-3**. As such, UE **101-1** may provide a session identifier, associated with the existing group communication session, to UE **101-4**. UE **101-1** may additionally provide information regarding blockchain system **109**, such as identifiers of one or more nodes **111** of blockchain system **109** (e.g., MDNs, IP addresses, etc. of UEs **101** that are serving as nodes **111** of blockchain system **109**). In situations where UE **101-4** is already an existing node **111** of blockchain system **109** (e.g., where UE **101-4** is "promoted" from a passive participant to an active participant of the group communication session), UE **101-1** may forgo providing (at **1902**) information regarding blockchain system **109** to UE **101-4**, as UE **101-4** may already be aware of blockchain system **109** associated with the group communication session.

In some embodiments, UE **101-1** may provide (at **1902**) the above-described information and/or other group communication session information to UE **101-4** via an RCS message or via some other suitable protocol or type of message. In some embodiments, UE **101-1** may generate an asymmetric key pair (e.g., including a public key and a private key), provide the public key to UE **101-4**, and encrypt some or all of the information (provided at **1902**) using the private key. In such embodiments, UE **101-4** may decrypt the information (provided at **1902**) using the public key.

As further shown, UE **101-4** may, based on being added (or promoted) to the group communication session, generate asymmetric key pair **103** (e.g., including a respective private key **105** and public key **107**), and may share public key **107** of the generated asymmetric key pair **103**. For example, UE **101-4** may record the generated (at **1904**) public key **107** to blockchain system **109** and/or to some other suitable device or system from which the other UEs **101-1**, **101-2**, and **101-3** of the group communication session may retrieve and/or validate the generated public key **107** associated with UE **101-4**.

UE **101-1** may also notify (at **1906**) the other participants of the group communication session (e.g., UEs **101-2** and **101-3**) that UE **101-4** has been added to the group communication session. The notification may include an identifier of UE **101-4**, such that UEs **101-2** and **101-3** may include UE **101-4** when sending encrypted messages via the group communication session.

Once UE 101-4 has been added to the group communication session, UE 101-4 may send encrypted messages to other participants of the group communication session (e.g., UEs 101-1, 101-2, and 101-3) in a similar manner as discussed above. For example, as shown in FIG. 20, UE 101-4 may obtain and/or cross-validate public keys 107-1, 107-2, and 107-3 in order to encrypt outgoing message 2001. As discussed above, encrypting message 2001 based on the obtained public keys 107-1, 107-2, and 107-3 may include generating an encrypted message package that includes multiple encrypted copies of message 2001, where each encrypted copy may be encrypted based on a respective public key 107 of a given recipient UE 101 and/or private key 105-4 of UE 101-4. UEs 101-1, 101-2, and 101-3 may obtain and/or cross-validate public key 107-4, associated with UE 101-4, in a manner similarly discussed above in order to decrypt such encrypted messages provided by UE 101-4.

FIG. 21 illustrates an example removal of a participant from a group communication session. As shown, UE 101-1 may output (at 2102) an indication to some or all of the other UEs 101 of the group communication session (e.g., UEs 101-2, 101-3, and 101-4, in this example) that UE 101-4 is being removed from the group communication session. The indication may include a session identifier associated with the group communication session and/or other suitable information based on which UEs 101-2, 101-3, and/or 101-4 may identify the group communication session from which UE 101-4 is being removed.

Based on receiving such indication, UE 101-4 may remove (at 2104) all keys associated with the group communication session. For example, UE 101-4 may remove, delete, mark for garbage collection, etc. private key 105-4, public key 107-4, and/or one or more public keys 107 associated with UEs 101-1, 101-2, and 101-3. In some embodiments, UE 101-4 may retain (at 2104) a chat history associated with the group communication session. For example, a messaging application of UE 101-4 may retain copies of messages (e.g., decrypted copies of such messages) received by UE 101-4 via the group communication session. UE 101-4 may retain such copies along with a session identifier of the group communication session. In this manner, a user of UE 101-4 may refer back to the previously received messages, even if UE 101-4 does not receive new messages via the group communication session after UE 101-4 is removed. Further, in instances where UE 101-4 is added back to the group communication session (e.g., associated with the same session identifier), a user interface, messaging application, etc. of UE 101-4 may seamlessly present newly received messages (e.g., after UE 101-4 is added back to the group communication session) along with previously received messages (e.g., received before UE 101-4 was removed from the group communication session).

UEs 101-2 and 101-3 may further remove (at 2106 and 2108, respectively) keys associated with UE 101-4, such as public key 107-4, based on receiving (at 2102) the indication that UE 101-4 has been removed. Additionally, UE 101-1 may remove (at 2110) keys associated with UE 101-4, such as public key 107-4, based on the removal of UE 101-4.

In situations where the removed UE 101-4 is serving as a node 111 of blockchain system 109, UE 101-4 may be “demoted” from an active participant of the group communication session to a passive participant. For example, UE 101-4 may continue serving as a node 111 of blockchain system 109, but may no longer receive encrypted messages via the group communication session from other UEs 101 of

the group communication session. In such examples, UE 101-4 may retain some or all public keys 107 associated with UEs 101-1, 101-2, and 101-3, at least inasmuch as these retained public keys 107 may be stored within a local copy of blockchain system 109 maintained by UE 101-4.

FIG. 22 illustrates an example environment 2200, in which one or more embodiments may be implemented. In some embodiments, environment 2200 may correspond to a Fifth Generation (“5G”) network, and/or may include elements of a 5G network. In some embodiments, environment 2200 may correspond to a 5G Non-Standalone (“NSA”) architecture, in which a 5G radio access technology (“RAT”) may be used in conjunction with one or more other RATs (e.g., a Long-Term Evolution (“LTE”) RAT), and/or in which elements of a 5G core network may be implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core (“EPC”)). As shown, environment 2200 may include UE 101, RAN 2210 (which may include one or more Next Generation Node Bs (“gNBs”) 2211), RAN 2212 (which may include one or more evolved Node Bs (“eNBs”) 2213), and various network functions such as Access and Mobility Management Function (“AMF”) 2215, Mobility Management Entity (“MME”) 2216, Serving Gateway (“SGW”) 2217, Session Management Function (“SMF”) / Packet Data Network (“PDN”) Gateway (“PGW”) - Control plane function (“PGW-C”) 2220, Policy Control Function (“PCF”) / Policy Charging and Rules Function (“PCRF”) 2225, Application Function (“AF”) 2230, User Plane Function (“UPF”) / PGW-User plane function (“PGW-U”) 2235, Unified Data Management function (“UDM”) / Home Subscriber Server (“HSS”) 2240, and Authentication Server Function (“AUSF”) 2245. Environment 2200 may also include one or more networks, such as Data Network (“DN”) 2250.

The example shown in FIG. 22 illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C 2220, PCF/PCRF 2225, UPF/PGW-U 2235, UDM/HSS 2240, and/or AUSF 2245). In practice, environment 2200 may include multiple instances of such components or functions. For example, in some embodiments, environment 2200 may include multiple “slices” of a core network, where each slice includes a discrete set of network functions (e.g., one slice may include a first instance of SMF/PGW-C 2220, PCF/PCRF 2225, UPF/PGW-U 2235, UDM/HSS 2240, and/or AUSF 2245, while another slice may include a second instance of SMF/PGW-C 2220, PCF/PCRF 2225, UPF/PGW-U 2235, UDM/HSS 2240, and/or AUSF 2245). The different slices may provide differentiated levels of service, such as service in accordance with different Quality of Service (“QoS”) parameters.

The quantity of devices and/or networks, illustrated in FIG. 22, is provided for explanatory purposes only. In practice, environment 2200 may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. 22. For example, while not shown, environment 2200 may include devices that facilitate or enable communication between various components shown in environment 2200, such as routers, modems, gateways, switches, hubs, etc. Alternatively, or additionally, one or more of the devices of environment 2200 may perform one or more network functions described as being performed by another one or more of the devices of environment 2200. Devices of environment 2200 may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of

wired and wireless connections. In some implementations, one or more devices of environment 2200 may be physically integrated in, and/or may be physically attached to, one or more other devices of environment 2200.

UE 101 may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN 2210, RAN 2212, and/or DN 2250. UE 101 may be, or may include, a radiotelephone, a personal communications system ("PCS") terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant ("PDA") (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a tablet computer, a camera, a personal gaming system, an IoT device (e.g., a sensor, a smart home appliance, or the like), a wearable device, an Internet of Things ("IoT") device, a Machine-to-Machine ("M2M") device, or another type of mobile computation and communication device. UE 101 may send traffic to and/or receive traffic (e.g., user plane traffic) from DN 2250 via RAN 2210, RAN 2212, and/or UPF/PGW-U 2235. In some embodiments, UE 101 may include, implement, etc. one or more nodes 111 of blockchain system 109.

RAN 2210 may be, or may include, a 5G RAN that includes one or more base stations (e.g., one or more gNBs 2211), via which UE 101 may communicate with one or more other elements of environment 2200. UE 101 may communicate with RAN 2210 via an air interface (e.g., as provided by gNB 2211). For instance, RAN 2210 may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE 101 via the air interface, and may communicate the traffic to UPF/PGW-U 2235, and/or one or more other devices or networks. Similarly, RAN 2210 may receive traffic intended for UE 101 (e.g., from UPF/PGW-U 2235, AMF 2215, and/or one or more other devices or networks) and may communicate the traffic to UE 101 via the air interface.

RAN 2212 may be, or may include, a LTE RAN that includes one or more base stations (e.g., one or more eNBs 2213), via which UE 101 may communicate with one or more other elements of environment 2200. UE 101 may communicate with RAN 2212 via an air interface (e.g., as provided by eNB 2213). For instance, RAN 2210 may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE 101 via the air interface, and may communicate the traffic to UPF/PGW-U 2235, and/or one or more other devices or networks. Similarly, RAN 2210 may receive traffic intended for UE 101 (e.g., from UPF/PGW-U 2235, SGW 2217, and/or one or more other devices or networks) and may communicate the traffic to UE 101 via the air interface.

AMF 2215 may include one or more devices, systems, Virtualized Network Functions ("VNFs"), etc., that perform operations to register UE 101 with the 5G network, to establish bearer channels associated with a session with UE 101, to hand off UE 101 from the 5G network to another network, to hand off UE 101 from the other network to the 5G network, manage mobility of UE 101 between RANs 2210 and/or gNBs 2211, and/or to perform other operations. In some embodiments, the 5G network may include multiple AMFs 2215, which communicate with each other via the N14 interface (denoted in FIG. 22 by the line marked "N14" originating and terminating at AMF 2215).

MME 2216 may include one or more devices, systems, VNFs, etc., that perform operations to register UE 101 with the EPC, to establish bearer channels associated with a

session with UE 101, to hand off UE 101 from the EPC to another network, to hand off UE 101 from another network to the EPC, manage mobility of UE 101 between RANs 2212 and/or eNBs 2213, and/or to perform other operations.

SGW 2217 may include one or more devices, systems, VNFs, etc., that aggregate traffic received from one or more eNBs 2213 and send the aggregated traffic to an external network or device via UPF/PGW-U 2235. Additionally, SGW 2217 may aggregate traffic received from one or more UPF/PGW-Us 2235 and may send the aggregated traffic to one or more eNBs 2213. SGW 2217 may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs 2210 and 2212).

SMF/PGW-C 2220 may include one or more devices, systems, VNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C 2220 may, for example, facilitate the establishment of communication sessions on behalf of UE 101. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF 2225.

PCF/PCRF 2225 may include one or more devices, systems, VNFs, etc., that aggregate information to and from the 5G network and/or other sources. PCF/PCRF 2225 may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF 2225).

AF 2230 may include one or more devices, systems, VNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

UPF/PGW-U 2235 may include one or more devices, systems, VNFs, etc., that receive, store, and/or provide data (e.g., user plane data). For example, UPF/PGW-U 2235 may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE 101, from DN 2250, and may forward the user plane data toward UE 101 (e.g., via RAN 2210, SMF/PGW-C 2220, and/or one or more other devices). In some embodiments, multiple UPFs 2235 may be deployed (e.g., in different geographical locations), and the delivery of content to UE 101 may be coordinated via the N9 interface (e.g., as denoted in FIG. 22 by the line marked "N9" originating and terminating at UPF/PGW-U 2235). Similarly, UPF/PGW-U 2235 may receive traffic from UE 101 (e.g., via RAN 2210, SMF/PGW-C 2220, and/or one or more other devices), and may forward the traffic toward DN 2250. In some embodiments, UPF/PGW-U 2235 may communicate (e.g., via the N4 interface) with SMF/PGW-C 2220, regarding user plane data processed by UPF/PGW-U 2235.

UDM/HSS 2240 and AUSF 2245 may include one or more devices, systems, VNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF 2245 and/or UDM/HSS 2240, profile information associated with a subscriber. AUSF 2245 and/or UDM/HSS 2240 may perform authentication, authorization, and/or accounting operations associated with the subscriber and/or a communication session with UE 101.

DN 2250 may include one or more wired and/or wireless networks. For example, DN 2250 may include an Internet Protocol ("IP")-based PDN, a wide area network ("WAN") such as the Internet, a private enterprise network, and/or one or more other networks. UE 101 may communicate, through DN 2250, with data servers, other UEs 101, and/or to other servers or applications that are coupled to DN 2250. DN

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2250 may be connected to one or more other networks, such as a public switched telephone network (“PSTN”), a public land mobile network (“PLMN”), and/or another network. DN 2250 may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE 101 may communicate.

FIG. 23 illustrates an example Distributed Unit (“DU”) network 2300, which may be included in and/or implemented by one or more RANs (e.g., RAN 2210, RAN 2212, or some other RAN). In some embodiments, a particular RAN may include one DU network 2300. In some embodiments, a particular RAN may include multiple DU networks 2300. In some embodiments, DU network 2300 may correspond to a particular gNB 2211 of a 5G RAN (e.g., RAN 2210). In some embodiments, DU network 2300 may correspond to multiple gNBs 2211. In some embodiments, DU network 2300 may correspond to one or more other types of base stations of one or more other types of RANs. As shown, DU network 2300 may include Centralized Unit (“CU”) 2305, one or more Distributed Units (“DUs”) 2303-1 through 2303-N (referred to individually as “DU 2303,” or collectively as “DUs 2303”), and one or more Radio Units (“RUs”) 2301-1 through 2301-M (referred to individually as “RU 2301,” or collectively as “RUs 2301”).

CU 2305 may communicate with a core of a wireless network (e.g., may communicate with one or more of the devices or systems described above with respect to FIG. 22, such as AMF 2215 and/or UPF/PGW-U 2235). In the uplink direction (e.g., for traffic from UEs 101 to a core network), CU 2305 may aggregate traffic from DUs 2303, and forward the aggregated traffic to the core network. In some embodiments, CU 2305 may receive traffic according to a given protocol (e.g., Radio Link Control (“RLC”)) from DUs 2303, and may perform higher-layer processing (e.g., may aggregate/process RLC packets and generate Packet Data Convergence Protocol (“PDCP”) packets based on the RLC packets) on the traffic received from DUs 2303.

In accordance with some embodiments, CU 2305 may receive downlink traffic (e.g., traffic from the core network) for a particular UE 101, and may determine which DU(s) 2303 should receive the downlink traffic. DU 2303 may include one or more devices that transmit traffic between a core network (e.g., via CU 2305) and UE 101 (e.g., via a respective RU 2301). DU 2303 may, for example, receive traffic from RU 2301 at a first layer (e.g., physical (“PHY”) layer traffic, or lower PHY layer traffic), and may process/aggregate the traffic to a second layer (e.g., upper PHY and/or RLC). DU 2303 may receive traffic from CU 2305 at the second layer, may process the traffic to the first layer, and provide the processed traffic to a respective RU 2301 for transmission to UE 101.

RU 2301 may include hardware circuitry (e.g., one or more RF transceivers, antennas, radios, and/or other suitable hardware) to communicate wirelessly (e.g., via an RF interface) with one or more UEs 101, one or more other DUs 2303 (e.g., via RUs 2301 associated with DUs 2303), and/or any other suitable type of device. In the uplink direction, RU 2301 may receive traffic from UE 101 and/or another DU 2303 via the RF interface and may provide the traffic to DU 2303. In the downlink direction, RU 2301 may receive traffic from DU 2303, and may provide the traffic to UE 101 and/or another DU 2303.

RUs 2301 may, in some embodiments, be communicatively coupled to one or more Multi-Access/Mobile Edge Computing (“MEC”) devices, referred to sometimes herein simply as “MECs” 2307. For example, RU 2301-1 may be communicatively coupled to MEC 2307-1, RU 2301-M may

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be communicatively coupled to MEC 2307-M, DU 2303-1 may be communicatively coupled to MEC 2307-2, DU 2303-N may be communicatively coupled to MEC 2307-N, CU 2305 may be communicatively coupled to MEC 2307-3, and so on. MECs 2307 may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE 101, via a respective RU 2301.

For example, RU 2301-1 may route some traffic, from UE 101, to MEC 2307-1 instead of to a core network (e.g., via DU 2303 and CU 2305). MEC 2307-1 may process the traffic, perform one or more computations based on the received traffic, and may provide traffic to UE 101 via RU 2301-1. In this manner, ultra-low latency services may be provided to UE 101, as traffic does not need to traverse DU 2303, CU 2305, and an intervening backhaul network between DU network 2300 and the core network. In some embodiments, MEC 2307 may include, and/or may implement, some or all of the functionality described above with respect to one or more nodes 111.

FIG. 24 illustrates example components of device 2400. One or more of the devices described above may include one or more devices 2400. Device 2400 may include bus 2410, processor 2420, memory 2430, input component 2440, output component 2450, and communication interface 2460. In another implementation, device 2400 may include additional, fewer, different, or differently arranged components.

Bus 2410 may include one or more communication paths that permit communication among the components of device 2400. Processor 2420 may include a processor, microprocessor, or processing logic that may interpret and execute instructions. In some embodiments, processor 2420 may be or may include one or more hardware processors. Memory 2430 may include any type of dynamic storage device that may store information and instructions for execution by processor 2420, and/or any type of non-volatile storage device that may store information for use by processor 2420.

Input component 2440 may include a mechanism that permits an operator to input information to device 2400 and/or other receives or detects input from a source external to 2440, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component 2440 may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System (“GPS”)-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component 2450 may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more light emitting diodes (“LEDs”), etc.

Communication interface 2460 may include any transceiver-like mechanism that enables device 2400 to communicate with other devices and/or systems. For example, communication interface 2460 may include an Ethernet interface, an optical interface, a coaxial interface, or the like. Communication interface 2460 may include a wireless communication device, such as an infrared (“IR”) receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device 2400 may include more than

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one communication interface **2460**. For instance, device **2400** may include an optical interface and an Ethernet interface.

Device **2400** may perform certain operations relating to one or more processes described above. Device **2400** may perform these operations in response to processor **2420** executing software instructions stored in a computer-readable medium, such as memory **2430**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical memory devices. The software instructions may be read into memory **2430** from another computer-readable medium or from another device. The software instructions stored in memory **2430** may cause processor **2420** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. 1-21), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed

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by multiple devices. Further, multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

To the extent the aforementioned implementations collect, store, or employ personal information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

What is claimed is:

1. A first device, comprising:

one or more hardware processors configured to:

- maintain an asymmetric key pair that includes a first key and a second key, wherein the asymmetric key pair is associated with a particular key identifier;
- receive a message to be sent to a second device and a third device;
- receive a first public key associated with the second device;
- receive a second public key associated with the third device;
- generate an encrypted communication based on the message, wherein generating the encrypted communication includes:
 - generating a first encrypted version of the message by encrypting the message using the first key of the asymmetric key pair and the first public key associated with the second device, and
 - generating a second encrypted version of the message by encrypting the message using the first key of the asymmetric key pair and the second public key associated with the third device,
 wherein the encrypted communication includes the particular key identifier, the first encrypted version of the message, and the second encrypted version of the message;
- output the encrypted communication to the second device and the third device; and
- output, to a blockchain system, the second key and the particular key identifier associated with the asymmetric key pair,

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wherein the second device:

obtains the second key from the blockchain system based on the particular key identifier included in the encrypted communication, and

decrypts the first encrypted version of the message using the second key obtained from the blockchain system based on the particular key identifier and a first private key that is associated with the first public key, and wherein the third device:

obtains the second key from the blockchain system based on the particular key identifier included in the encrypted communication, and

decrypts the second encrypted version of the message using the second key obtained from the blockchain system based on the particular key identifier and a second private key that is associated with the second public key.

2. The first device of claim 1, wherein receiving the first and second public keys includes obtaining the first and second public keys from the blockchain system.

3. The first device of claim 1, wherein the one or more hardware processors are further configured to:

add a fourth device to a group communication session with which the first, second, and third devices are associated,

wherein adding the fourth device includes providing a session identifier associated with the group communication session to the fourth device, wherein the fourth device obtains at least of the first or second public keys from the blockchain system based on the session identifier.

4. The first device of claim 1, wherein the one or more hardware processors are further configured to:

remove the third device from a group communication session with which the first, second, and third devices are associated,

wherein removing the third device from the group communication session includes outputting an indication to the second and third devices that the third device is being removed from the group communication session, wherein the third device retains a chat history associated with the group communication session after receiving the indication that the third device is being removed from the group communication session.

5. The first device of claim 1, wherein the second device is a node of the blockchain system, wherein obtaining the second key from the blockchain system includes cross-validating the second key with at least a threshold quantity of other nodes of the blockchain system.

6. The first device of claim 1, wherein the one or more hardware processors are further configured to:

remove the third device from a group communication session with which the first, second, and third devices are associated,

wherein removing the third device from the group communication session includes outputting an indication to the second device that the third device is being removed from the group communication session, wherein the second device deletes the second public key associated with the third device.

7. The first device of claim 1, wherein the third device is a node of the blockchain system, wherein the one or more hardware processors are further configured to:

remove the third device from a group communication session with which the first, second, and third devices are associated,

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wherein the third device continues to serve as a node of the blockchain system after being removed from the group communication session.

8. A non-transitory computer-readable medium storing a plurality of processor-executable instructions, wherein execution of the processor-executable instructions by a first device causes the first device to:

maintain an asymmetric key pair that includes a first key and a second key, wherein the asymmetric key pair is associated with a particular key identifier;

receive a message to be sent to a second device and a third device;

receive a first public key associated with the second device;

receive a second public key associated with the third device;

generate an encrypted communication based on the message, wherein generating the encrypted communication includes:

encrypting the message using the first key of the asymmetric key pair and the first public key associated with the second device, and

generating a second encrypted version of the message by encrypting the message using the first key of the asymmetric key pair and the second public key associated with the third device,

wherein the encrypted communication includes the particular key identifier, the first encrypted version of the message, and the second encrypted version of the message;

output the encrypted communication to the second device and the third device; and

output, to a blockchain system, the second key and the particular key identifier associated with the asymmetric key pair,

wherein the second device:

obtains the second key from the blockchain system based on the particular key identifier included in the encrypted communication, and

decrypts the first encrypted version of the message using the second key obtained from the blockchain system based on the particular key identifier and a first private key that is associated with the first public key, and

wherein the third device:

obtains the second key from the blockchain system based on the particular key identifier included in the encrypted communication, and

decrypts the second encrypted version of the message using the second key obtained from the blockchain system based on the particular key identifier and a second private key that is associated with the second public key.

9. The non-transitory computer-readable medium of claim 8, wherein receiving the first and second public keys includes obtaining the first and second public keys from the blockchain system.

10. The non-transitory computer-readable medium of claim 8, wherein the plurality of processor-executable instructions further include processor-executable instructions to:

add a fourth device to a group communication session with which the first, second, and third devices are associated,

wherein adding the fourth device includes providing a session identifier associated with the group communication session to the fourth device, wherein the fourth

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device obtains at least of the first or second public keys from the blockchain system based on the session identifier.

11. The non-transitory computer-readable medium of claim 8, wherein the plurality of processor-executable instructions further include processor-executable instructions to:

remove the third device from a group communication session with which the first, second, and third devices are associated,

wherein removing the third device from the group communication session includes outputting an indication to the second and third devices that the third device is being removed from the group communication session, wherein the third device retains a chat history associated with the group communication session after receiving the indication that the third device is being removed from the group communication session.

12. The non-transitory computer-readable medium of claim 8, wherein the second device is a node of the blockchain system, wherein obtaining the second key from the blockchain system includes cross-validating the second key with at least a threshold quantity of other nodes of the blockchain system.

13. The non-transitory computer-readable medium of claim 8, wherein the plurality of processor-executable instructions further include processor-executable instructions to:

remove the third device from a group communication session with which the first, second, and third devices are associated,

wherein the third device continues to serve as a node of the blockchain system after being removed from the group communication session.

14. A method, comprising:

maintaining, by a first device, an asymmetric key pair that includes a first key and a second key, wherein the asymmetric key pair is associated with a particular key identifier;

receiving a message to be sent to a second device and a third device;

receiving a first public key associated with the second device;

receiving a second public key associated with the third device;

generating an encrypted communication based on the message, wherein generating the encrypted communication includes:

encrypting the message using the first key of the asymmetric key pair and the first public key associated with the second device, and

generating a second encrypted version of the message by encrypting the message using the first key of the asymmetric key pair and the second public key associated with the third device,

wherein the encrypted communication includes the particular key identifier, the first encrypted version of the message, and the second encrypted version of the message;

outputting the encrypted communication to the second device and the third device; and

outputting, to a blockchain system, the second key and the particular key identifier associated with the asymmetric key pair,

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wherein the second device:

obtains the second key from the blockchain system based on the particular key identifier included in the encrypted communication, and

decrypts the first encrypted version of the message using the second key obtained from the blockchain system based on the particular key identifier and a first private key that is associated with the first public key, and

wherein the third device:

obtains the second key from the blockchain system based on the particular key identifier included in the encrypted communication, and

decrypts the second encrypted version of the message using the second key obtained from the blockchain system based on the particular key identifier and a second private key that is associated with the second public key.

15. The method of claim 14, wherein receiving the first and second public keys includes obtaining the first and second public keys from the blockchain system.

16. The method of claim 14, further comprising:

adding a fourth device to a group communication session with which the first, second, and third devices are associated,

wherein adding the fourth device includes providing a session identifier associated with the group communication session to the fourth device, wherein the fourth device obtains at least of the first or second public keys from the blockchain system based on the session identifier.

17. The method of claim 14, further comprising:

removing the third device from a group communication session with which the first, second, and third devices are associated,

wherein removing the third device from the group communication session includes outputting an indication to the second and third devices that the third device is being removed from the group communication session, wherein the third device retains a chat history associated with the group communication session after receiving the indication that the third device is being removed from the group communication session.

18. The method of claim 14, wherein the second device is a node of the blockchain system, wherein obtaining the second key from the blockchain system includes cross-validating the second key with at least a threshold quantity of other nodes of the blockchain system.

19. The method of claim 14, further comprising:

removing the third device from a group communication session with which the first, second, and third devices are associated,

wherein removing the third device from the group communication session includes outputting an indication to the second device that the third device is being removed from the group communication session, wherein the second device deletes the second public key associated with the third device.

20. The method of claim 14, wherein the third device is a node of the blockchain system, the method further comprising:

removing the third device from a group communication session with which the first, second, and third devices are associated,

wherein the third device continues to serve as a node of the blockchain system after being removed from the group communication session.

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